

FCE 201: THEORY OF STRUCTURES I A

INTRODUCTORY REMARKS

The field of **Civil Engineering** includes: **Structural Engineering**, Water Resources Engineering, Public Health Engineering, Transportation Engineering and Geotechnical Engineering

Structural Engineering: -

1. **Theory of structures** (2nd, 3rd, and 4th year = 6 course units)
2. Strength of Materials (2nd, and 3rd year = 4 course units)
3. Civil Eng Materials (2nd and 3rd year = 4 course units)
4. Engineering Mathematics (2nd and 3rd year = 4 course units)
5. Engineering Drawing (2nd, and 3rd year = 2 course units)
6. **Structural Design** (4th and 5th year = 4 course units)

Relative “difficulty”: ToS requires 90% of your study time.

Recommendations: -

1. Study timetable: Mon, Tue, Wed, Thu, Fri = ToS; Sat = ALL others, Sun = Relax.
2. Punctuality: Lectures = 9:00 am prompt. No excuse at ALL.
3. Cell phones: Not allowed in lecture room: keep them off or silent or on vibrate mode.
4. Group work is better than lone ranger tactics.

Method of teaching: -

Lectures by dictation + written notes + worked examples + handouts + tutorial questions

Assessment: Two CATs (30%) + End of Semester Exam (70%)

Course Outline

0. Introduction
1. Concepts and Principles
2. **The Cable**
3. The Arch
4. The Truss
5. Beams and Frames

CHAPTER 0: INTRODUCTION

0.1 KINDS OF STRUCTURES

There are two kinds of structures namely:-

1. **Natural structures**

These are structures which occur in nature (designed and constructed by God).

Examples include **Skeleton** of an animal, **Trunk** of a tree, **Shell** of a tortoise or **Web** of a spider

2. **Artificial structures**

These are structures which are designed and constructed by man or woman.

Examples include: buildings, bridges, dams, tunnels, silos, stadia, water tanks, etc

Other examples include: wheelbarrows, bicycles, motor vehicles ships, aircraft, etc

Civil Engineering Structures do NOT move

Mechanical Engineering Structures move

0.2 GENERAL DEFINITION OF A STRUCTURE

A Structure may be generally defined as anything that:-

- Either, supports or resists forces of some kind
- Or, acts as a protective cover for something or someone
- Or, does both of the above

0.3 STRUCTURAL FORMS

Structures may be divided into two categories depending on their structural form:-

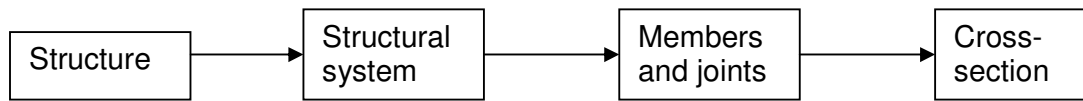
1. **Skeletal structures:** These are structures which are made up of distinct, ideally linear members. Examples of skeletal structures include ALL frameworks (trusses, grills, frames and arches). *[In FCE 201 we shall deal with skeletal structures only].*
2. **Continuum structures:** These are structures which are made up of members which can only be defined in at least two dimensions. Examples include a floor slab, a wall, and a dam.

0.4 STRUCTURAL FUNCTIONAL UNITS

A structure is considered to be composed of structural units as shown below:

- A structure is made up of several structural systems working together.
- Each structural system is made up of several members inter-connected and functioning in such a manner as to give the system complete structural independence.

- Each member is made up of a series of cross-sections from one end to the other.
- Each cross-section is seen to be suffering a specific level of the load effects.



Note: Compare with the functional units of an organism in biology:-

Organism → System → Organ → Tissue → Cell

0.5 STRUCTURAL ANALYSIS

When a load is applied to a structure, the structure will respond by developing stresses in its material. As a consequence of these stresses, the material strains and hence the structure deforms.

The aim of structural analysis is to **predict** (i.e. to determine in advance) the **internal stresses** that will be induced in the material of a structure due to a given set of loads, and the **deformations** that the structure will have to suffer. The internal stresses are called **member forces**, while the deformations are referred to as the **displacements**.

0.6 STRUCTURAL DESIGN

Structural design refers to all that engineering work (modeling, analysis, specifications, detailing) which results in creating on paper **how** what is proposed **must** be constructed on site, so that its structural behavior is **acceptable**.

Structural behavior is considered acceptable if the structure is **safe, stable and serviceable** throughout its design life.

Welcome to the
world of ToS

CHAPTER 1: CONCEPTS AND PRINCIPLES

1.1 CONCEPTS

[Concept = A notion retained in the mind, by reasoning or imagination]

1.1.1 A Structure

A **structure** is defined as an assembly of members **joined** together and **supported** in such a manner as to form a **stable framework**.

The framework must be able to **transmit** the **applied loads safely** from their **points of application**, via the **joints**, to the **points of restraint**, in a **stable** manner and without **excessive deformation**.

1.1.2 Structural Analysis

Structural analysis refers to the techniques used in order **to predict the behavior (response) of a structure** when subjected to applied loads. “**Theory of structures**” refers to a series of courses which deal with structural analysis.

In FCE 201 we shall **determine** (i.e predict in advance) the **external forces** (called **support reactions**) induced at the supports, and the **internal stresses** (called **member forces**) induced in the members due to a given set of loads acting on a given structure. In FCE 202 we shall **determine** the **displacements** caused by the applied loads.

1.1.3 Structural Idealization

Structural idealization is the **simplification** of an actual structural system to a model that can be analyzed easily using mathematical and physical principles. Such simplification is only **for purposes of structural analysis**.

The simplification is achieved by making appropriate assumptions regarding:-

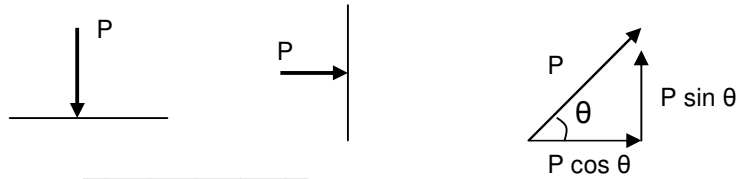
1. The nature of the applied loads.
2. The nature and behavior of the members.
3. The nature and behavior of the joints.
4. The nature and behavior of the points of restraint

Structural idealization results in what is termed the **idealized structure** or **structural model**.

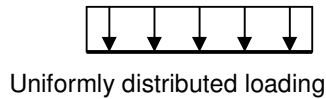
1.1.4 Idealization of Applied Loads

Within the scope of FCE 201, applied loads will be idealized into three categories:-

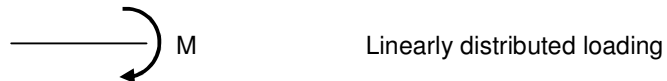
1. Point loads



2 Distributed loads



3 Bending moment



For purposes of analysis, all applied loads shall be considered with regard to the Cartesian coordinate system; and hence your knowledge of resolution of forces is a pre-requisite.

1.1.5 Idealization of Structural Members

In idealizing structural members, the following assumptions are made:-

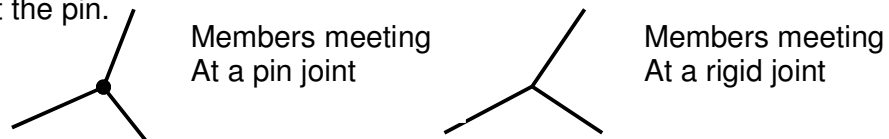
1. The member is **prismatic**, or is linearly tapered.
2. The member is linear, or curvilinear.
3. The material is elastic, isotropic, and homogeneous.
4. The material is NOT stressed beyond the elastic limit.
5. The member may be represented by a line coinciding with its longitudinal axis.
6. The self weight of the member is negligible.

1.1.6 Idealization of Joints

Joints are idealized into two categories:-

1. Pin joint (also called a hinge):

A pin joint can resist force in the x- and/or y-direction, but cannot resist any bending moment about the pin.

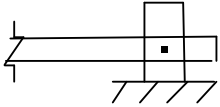
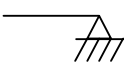
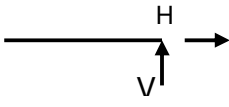
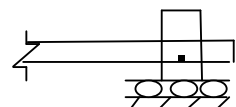
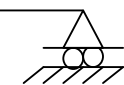
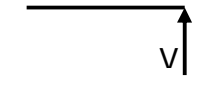
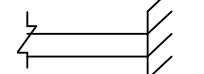
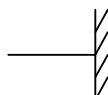
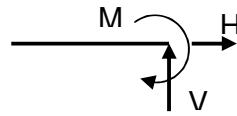


2. Rigid joint

A rigid joint can resist force in the x- and/or y-direction, and can resist bending moment.

1.1.7 Idealization of Supports (or points of restraint)

There are three basic types of supports as shown in the table below.

Real Model	Idealized Model	Free body diagram	Remarks
			Can resist a force in the x-direction Can resist a force in the y-direction Cannot resist moment about the z-axis
			Can resist a force in the y-direction Cannot resist a force in the x-direction Cannot resist moment about the z-axis
			Can resist a force in the x-direction Can resist a force in the y-direction Can resist moment about the z-axis

Note: The idealized loads, members, joints, and supports are only conceptual.

In practice it may be difficult to achieve ideal conditions.

1.1.8 Free Body Diagram

A “free body” is one that is **imagined** to be **freed** (i.e. totally disconnected) from all of its surrounding members, joints, or supports, whose presence is acknowledged only by the forces and moments they apply to the free body.

A free body may consist of part of a member, or a whole member, or several members, or the whole structure; BUT the free body must be able to sustain **all** the forces and moments acting on it in equilibrium, and in a stable manner.

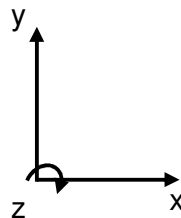
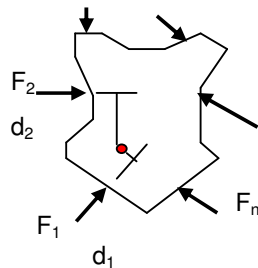
A sketch of a free body showing the body and **all** the forces and moments acting on it is referred to as a free body diagram.

1.2 PRINCIPLES

1.2.1 Principle of Static Equilibrium

The principle of static equilibrium states that, for a free body to remain **static**, the **algebraic sum** of all the **resolved forces** in any particular direction must equal zero; AND the **resultant moment** must also equal zero.

Consider the free body shown below (which is a 2-D system).



For a 3-D system, six equations of equilibrium must be satisfied.

$$\begin{aligned}\Sigma F_x &= 0 & \Sigma M_x &= 0 \\ \Sigma F_y &= 0 & \Sigma M_y &= 0 \\ \Sigma F_z &= 0 & \Sigma M_z &= 0\end{aligned}$$

The principle of **static equilibrium** requires that, for this free body to be in static equilibrium, three equations must be satisfied:-

$$\Sigma F_x = F_1 \cos \theta_1 + F_2 \cos \theta_2 + \dots + F_n \cos \theta_n = 0$$

$$\Sigma F_y = F_1 \sin \theta_1 + F_2 \sin \theta_2 + \dots + F_n \sin \theta_n = 0$$

$$\Sigma M_z = F_1 d_1 + F_2 d_2 + \dots + F_n d_n = 0$$

In these equations, θ_i is the inclination of the i^{th} force with respect to the x-axis.

Note that moments are taken about any convenient point on the plane of the body.

In this chapter we shall apply the principle of static equilibrium to determine the support reaction components induced at the supports of a structure due to a given set of loads.

1.2.2 Principle of Compatibility

The principle of compatibility states that the strains distribution over the whole structure must be **compatible** with the shape of the deformed structure. In simpler terms, the principle states that the structural form must show **continuity** both before and after straining.

1.2.3 Principle of Constitutivity

The principle of constitutivity states that the stresses must be related to the strains by means of the material **constitutive** properties. In simpler terms the principle states that the stresses are related to the strains by Hooke's law of linear elasticity. Thus $\sigma = E \epsilon$

1.2.4 Principle of Superposition

The principle of superposition states that the structural response due to a number of simultaneously applied loads is equal to the algebraic sum of the responses due to the individual loads each applied separately to the structure.

1.3 DETERMINATION OF SUPPORT REACTION COMPONENTS

The procedure for the determination of support reaction components is as follows:-

Step 1: Isolate and sketch an **appropriate** free body diagram.

[All the external forces and moments acting on it **must** be shown].

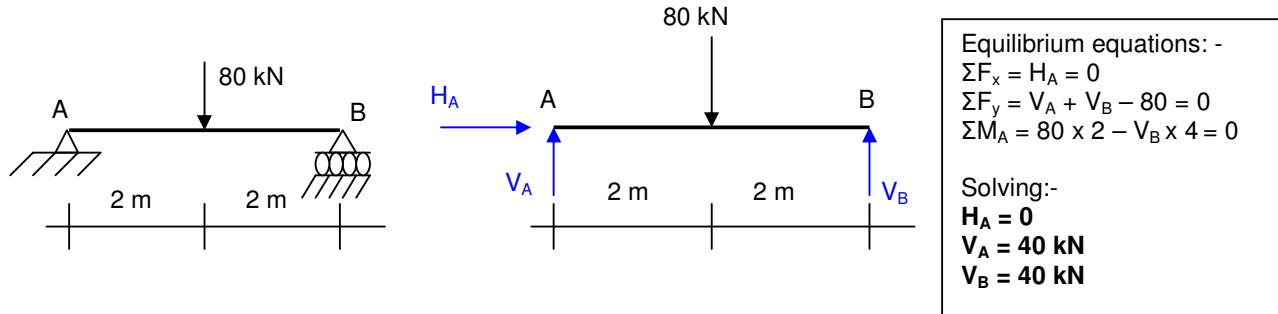
Step 2: **Set up** the equations of static equilibrium.

Step 3: Solve for the **unknown** forces.

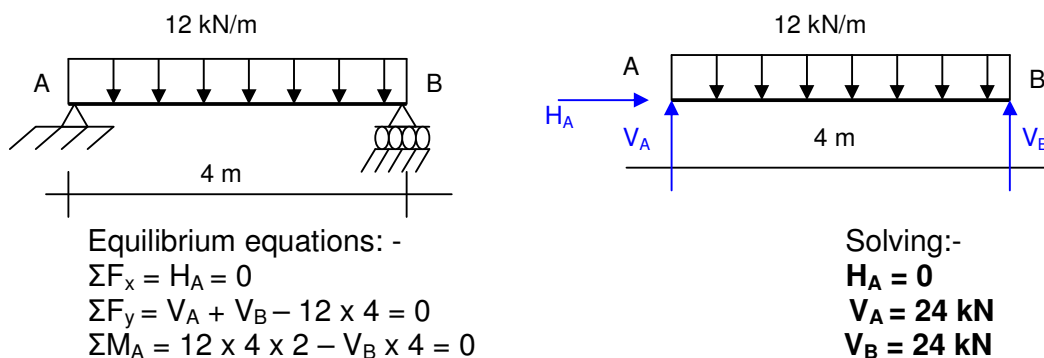
Worked Examples

Determine the **support reaction components** induced at the supports of each of the beam structures shown in examples 1 to 7.

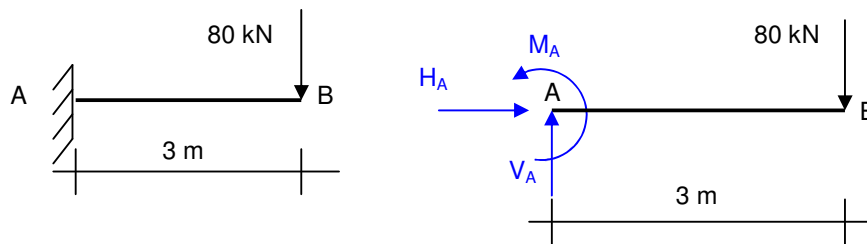
Example 1 (A simply supported beam carrying a central point load)



Example 2 (A simply supported beam carrying a u.d.l.)



Example 3 (A cantilever carrying a point load at the free end)



Equilibrium equations: -

$$\Sigma F_x = H_A = 0$$

$$\Sigma F_y = V_A - 80 = 0$$

$$\Sigma M_A = 80 \times 3 - M_A = 0$$

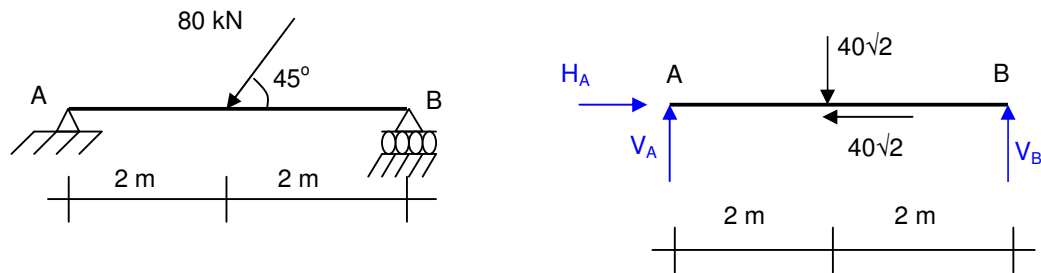
Solving:-

$$H_A = 0$$

$$V_A = 80 \text{ kN}$$

$$M_A = 240 \text{ kNm}$$

Example 4 (A simply supported beam carrying an inclined point load)



Equilibrium equations: -

$$\Sigma F_x = H_A - 40\sqrt{2} = 0$$

$$\Sigma F_y = V_A + V_B - 40\sqrt{2} = 0$$

$$\Sigma M_A = 40\sqrt{2} \times 2 - V_B \times 4 = 0$$

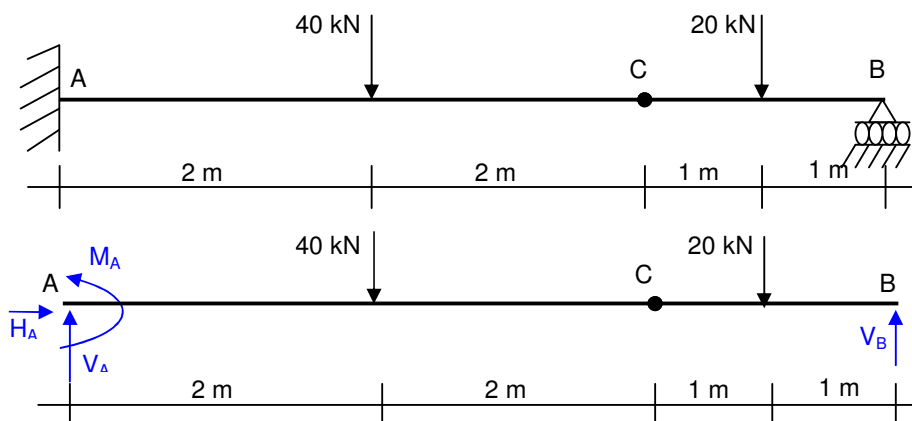
Solving:-

$$H_A = 40\sqrt{2} \text{ kN}$$

$$V_A = 20\sqrt{2} \text{ kN}$$

$$V_B = 20\sqrt{2} \text{ kN}$$

Example 5 (A propped cantilever with an internal hinge)



Equilibrium equations: -

$$\Sigma F_x = H_A = 0$$

$$\Sigma F_y = V_A + V_B - 40 - 20 = 0$$

$$\Sigma M_A = 40 \times 2 + 20 \times 5 - V_B \times 6 - M_A = 0$$

$$\Sigma M_{C(R)} = 20 \times 1 - V_B \times 2 = 0$$

Solving:-

$$H_A = 0$$

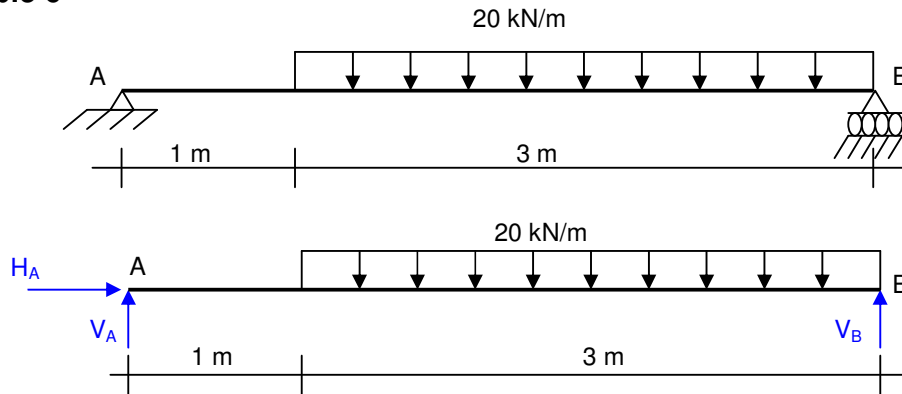
$$V_A = 50 \text{ kN}$$

$$V_B = 10 \text{ kN}$$

$$M_A = 120 \text{ kNm}$$

NOTE: The above solution can alternatively be achieved by isolating two free bodies AC and CB.

Example 6



Equilibrium equations: -

$$\Sigma F_x = H_A = 0$$

$$\Sigma F_y = V_A + V_B - 20 \times 3 = 0$$

$$\Sigma M_A = 20 \times 3 \times 2.5 - V_B \times 4 = 0$$

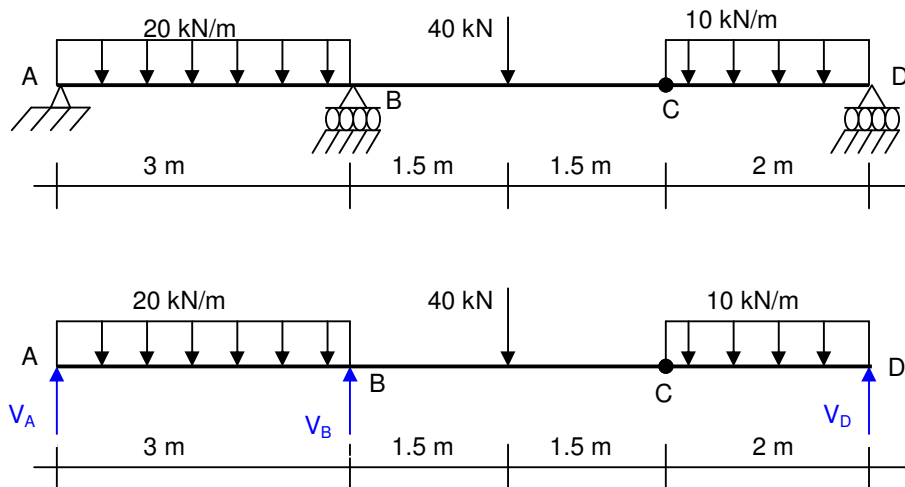
Solving:-

$$H_A = 0$$

$$V_A = 22.5 \text{ kN}$$

$$V_B = 37.5 \text{ kN}$$

Example 7



Equilibrium equations: -

$$\Sigma F_y = V_A + V_B + V_D - 20 \times 3 - 40 - 10 \times 2 = 0$$

$$\Sigma M_A = 20 \times 3 \times 1.5 + 40 \times 4.5 + 10 \times 2 \times 7 - 3 \times V_B - 8 \times V_D$$

$$\Sigma M_{C\odot} = 10 \times 2 \times 1 - 2 \times V_D = 0$$

Solving:-

$$V_A = 0$$

$$V_B = 110 \text{ kN}$$

$$V_D = 10 \text{ kN}$$

Note that so long as there are no applied loads with components in the x-direction, the horizontal reaction component H_A will turn out to be zero. Hence the equilibrium equation in the x-direction is not necessary.

1.4 STATIC INDETERMINACY

1.4.1 Degree of Static Indeterminacy (dsi)

1. A Structure is said to be **statically determinate** if all the external and internal force resultants can be determined by applying the equations of static equilibrium only.
2. If the equations of static equilibrium are not sufficient to determine all the external and internal force resultants, the structure is said to be **statically indeterminate**.
3. The number of additional equations needed to solve for all the unknown forces is termed the **degree of static indeterminacy (denoted by η)**. These additional equations are obtained by invoking the principles of **compatibility and constitutivity**.
4. In FCE 201 we shall deal with statically determinate structures only. However, we need to be able to distinguish between statically determinate and statically indeterminate structures. Such distinction is easily done by evaluating the degree of static indeterminacy:

1.4.2 Determination of dsi

For any given structure: -

1. Isolate appropriate fbd(s) and identify all unknown external and internal forces, **r**.
2. Determine the number of possible equations of equilibrium, **e**.
3. Evaluate the dsi **$\eta = r - e$**
4. Make deductions as follows: -
 - If **$r > e \Rightarrow \eta$ is +ve $\Rightarrow$ the structure is statically indeterminate**
 - If **$r = e \Rightarrow \eta = 0 \Rightarrow$ the structure is statically determinate**
 - If **$r < e \Rightarrow \eta$ is -ve $\Rightarrow$ the structure is **unstable!****

[In FCE 201 and FCE 202 we shall not deal with unstable structures]

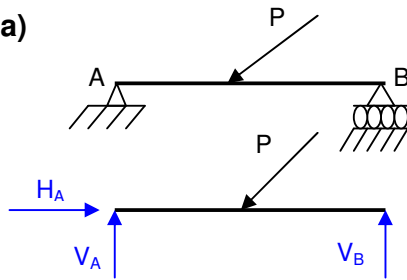
1.4.3 Geometric Instability

1. A **stable structure** is one which is restrained (i.e. supported) in such a manner that it responds to any applied loading by developing stresses instantaneously, and hence the structure **remains static** and does not change its original geometry appreciably.
2. An **unstable structure** is one which **undergoes rigid body motion** or changes its original geometry appreciably before developing any stresses. Such a structure is called a **mechanism** and is considered to suffer from **geometric instability**.
3. The principle of static equilibrium and the concept of degree of static indeterminacy are not applicable to a mechanism. However, the structural engineer must be able to recognize geometric instability so as to provide additional restraints to stabilize the structure.

Example 8

Determine the degree of static indeterminacy for each of the structures shown below.

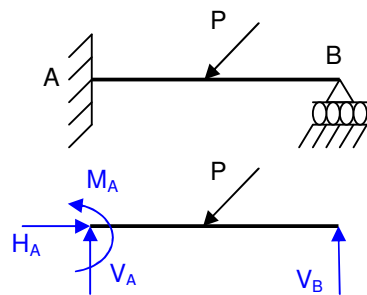
(a)



$$r = 3, e = 3, \text{ and } \eta = 0$$

Structure is **statically determinate**.

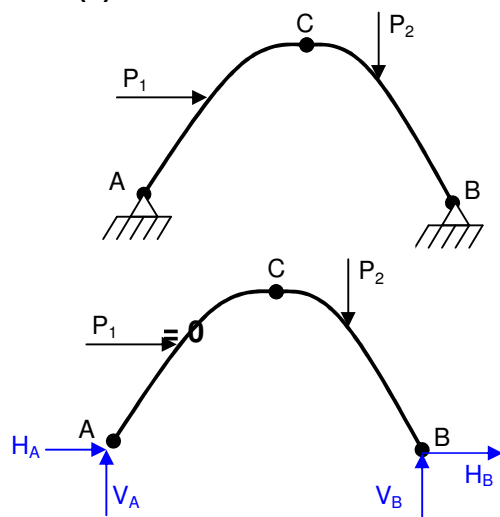
(c)



$$r = 4, e = 3 \text{ and } \eta = 1$$

Structure is **statically indeterminate to the first degree**

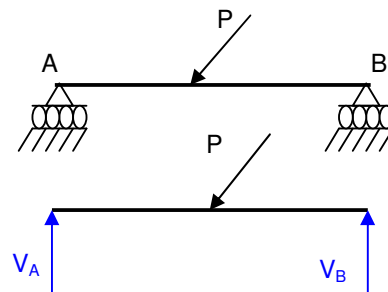
(e) ALT 1



$$r = 4; e = 3 + 1; \eta = 0$$

The structure is **statically determinate**

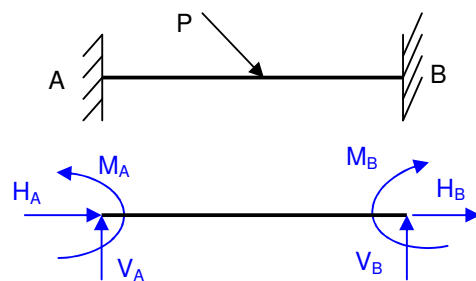
(b)



$$r = 2, e = 3 \text{ and } \eta = -1$$

Structure is **unstable**!

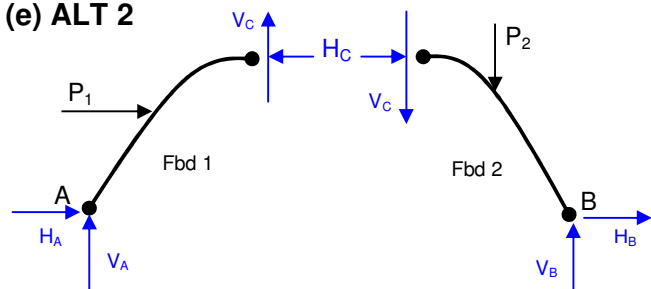
(d)



$$r = 6, e = 3 \text{ and } \eta = 3$$

Structure is **statically indeterminate to the third degree**

(e) ALT 2



$$\text{In this case: } r = 6; e = 3 + 3;$$

Note

1. The dsi is independent of the applied loads and/or the dimensions of the structure. The dsi depends only on the arrangement of supports, members and joints.
2. The choice of which fbd(s) to isolate is a matter of convenience other than principle.
3. When a "cut" is imagined through a member or joint The exact number of unknown internal forces and moments acting on that section must be revealed

1.5 SPECIAL CASES OF STATIC EQUILIBRIUM

Case 1

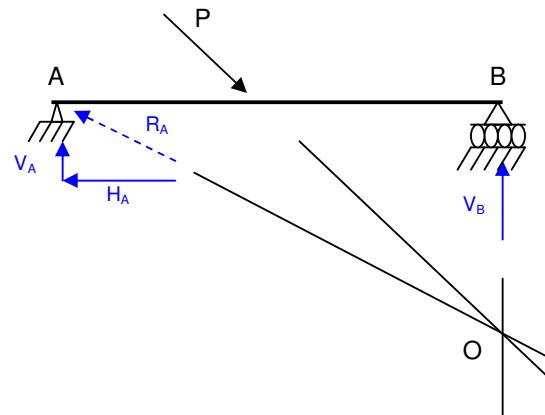
If there are **only two forces** acting on a plane body which is in static equilibrium, then the forces must be **equal** in magnitude, **opposite** in sense, and **collinear**.

Case 2

If there are **only three forces** acting on a plane body which is in static equilibrium, then the forces must be **concurrent** (i.e. they must all concur at some point). Consider the simply

supported beam shown below: -

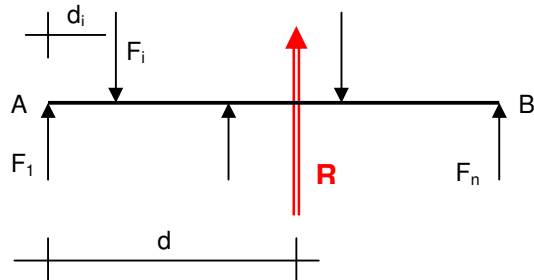
Since V_B must be vertical, and coincides with the line of action of P at the point O , then V_A and H_A must be such that their resultant R_A must pass through the point of concurrence, O .



Case 3

A set of **parallel forces** acting on a plane body can be reduced to the **resultant force** or **resultant couple**.

The algebraic sum of all the forces gives the resultant force: $R = \Sigma F_i$



Sum of moments about, say point A, gives the distance d , which defines the point of action of R : $Rd = \Sigma F_i d_i$

If $R = 0$, then **EITHER** the system is in equilibrium **OR** the system experiences a resultant couple. To determine which is which, sum moments about any point on the plane of the body, say point A.

If $\Sigma M_A = 0$ then the system is in **static equilibrium**

If $\Sigma M_A = M$, then M is the **resultant couple**.

1.6 OBJECTIVE AND SCOPE OF FCE 201

The objective of FCE 201 is to enable the student to analyze statically determinate structures using the concepts and principles laid down in this chapter, together with the specific techniques that shall be learnt in subsequent chapters.

Statically determinate structures can be divided into four categories. Each category shall constitute a chapter as shown in the course outline.

The analysis shall be limited to determining the **support reaction components** induced at the supports and the **internal member forces** induced at any cross-section of any member due to a given set of loads.

The deformations suffered by the cable shall be dealt with in FCE 201. The techniques for the determination of the deformations and displacements of beams, frames, trusses, hybrid frames and arches are dealt with in FCE 202.

Best of Luck Guys

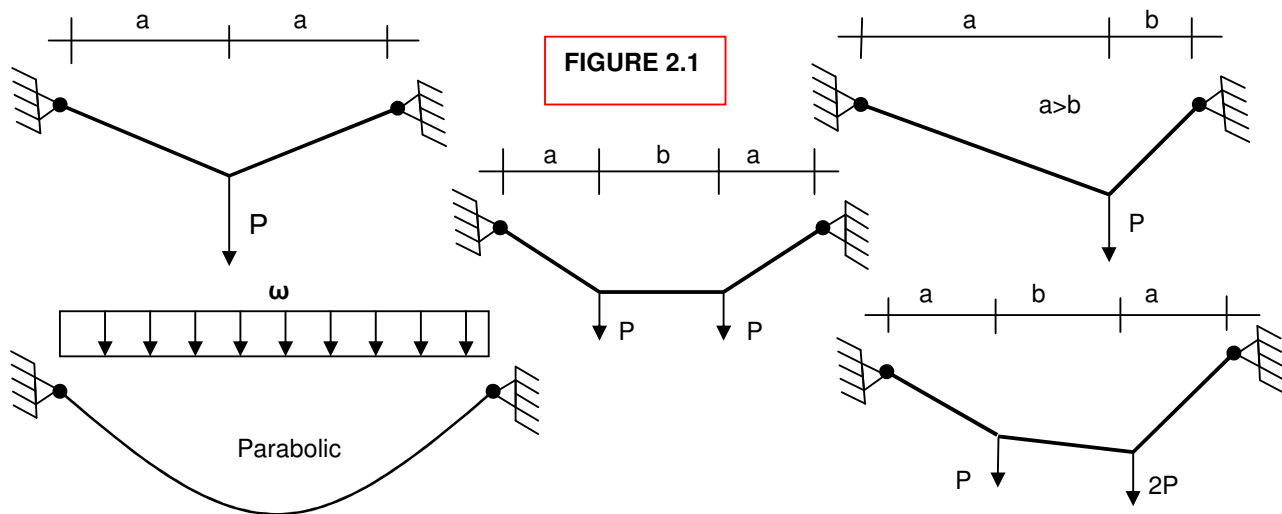
2.0 THE CABLE

2.1 INTRODUCTION

A cable transmits the applied loads to the supports by developing **axial tension** only. The cable is thus referred to as an **axial force structure**. The material used to make the cable must therefore be of high tensile strength.

Applications: Hangers (lifts, cranes, maintenance crabs, bridges); **Guys** (guyed communication masts); **Stays** (cable stayed bridges); **Cables** (suspension bridges, difficult construction sites).

The shape adopted by the cable is a function of the applied loads (i.e. the shape of the cable will depend on the nature of the loads, their magnitudes, and their points of application). The cable is therefore referred to as a **form-active structure**. It assumes the most efficient geometry for any given set of loads. Figure 2.1 illustrates the different shapes adopted by the same cable subjected to different loading.



The analysis of a cable for any given load scenario involves the determination of the support reaction components, the tensile force induced in the cable, the cable profile, the required cable length, and the expected cable elongation.

The analysis of a cable subjected to point loads relies directly on the principle of static equilibrium, some mathematics, and a bit of common sense.

However, the analysis of a cable subjected to a uniformly distributed load requires some good knowledge of calculus.

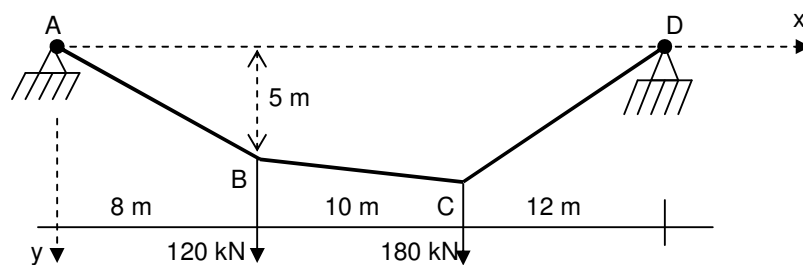
2.2 CABLE SUBJECTED TO POINT LOADS

The analysis of a cable subjected to point loads is illustrated in Example 1 below.

Example 1

Analyse the cable shown below and determine: -

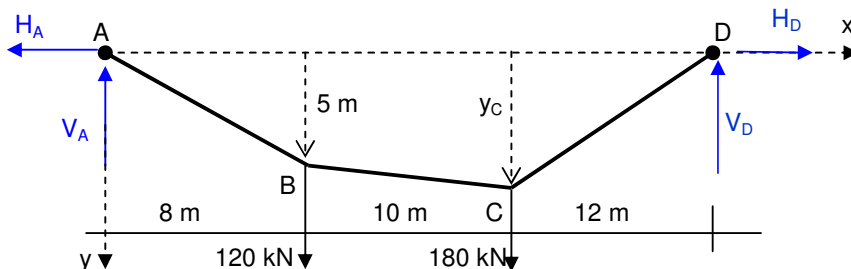
- The support reaction components at A and D.
- The maximum tension induced in the cable.
- The ordinate of the point C.
- The required cable length



Solution

(a) Support Reaction Components

Isolate the whole cable as a free body:



$$\Sigma F_x = -H_A + H_D = 0 \quad \text{----- (1)}$$

$$\Sigma F_y = V_A + V_D - 300 = 0 \quad \text{----- (2)}$$

$$\Sigma M_A = 120 \times 8 + 180 \times 18 - V_D \times 30 = 0 \quad \text{----- (3)}$$

The fourth equation is generated by considering the equilibrium of the portion AB

$$\Sigma M_{B(AB)} = V_A \times 8 - H_A \times 5 = 0 \quad \text{----- (4)}$$

Solving: - $V_A = 160 \text{ kN}$; $H_A = 256 \text{ kN}$; $V_D = 140 \text{ kN}$; $H_D = 256 \text{ kN}$.

(b) Maximum Tension

Since H is constant, the maximum tension occurs when the vertical component V attains its highest value. This will be in the portion AB where $V_A = 160 \text{ kN}$.

$$\text{Hence } T_{\max} = T_{AB} = \sqrt{(H_A^2 + V_A^2)} = \sqrt{(256^2 + 160^2)} = 301.9 \text{ kN}$$

(c) Ordinate of Point C

Isolate the portion C to D,

$$\Sigma M_{C(D)} = H_D \times y_C - V_D \times 12 = 0 \Rightarrow 256y_C = 12(140) \Rightarrow y_C = 6.563 \text{ m}$$

(d) Cable Length

By Pythagoras theorem:

$$L_{AB} = \sqrt{8^2 + 5^2} = 9.434 \text{ m}$$

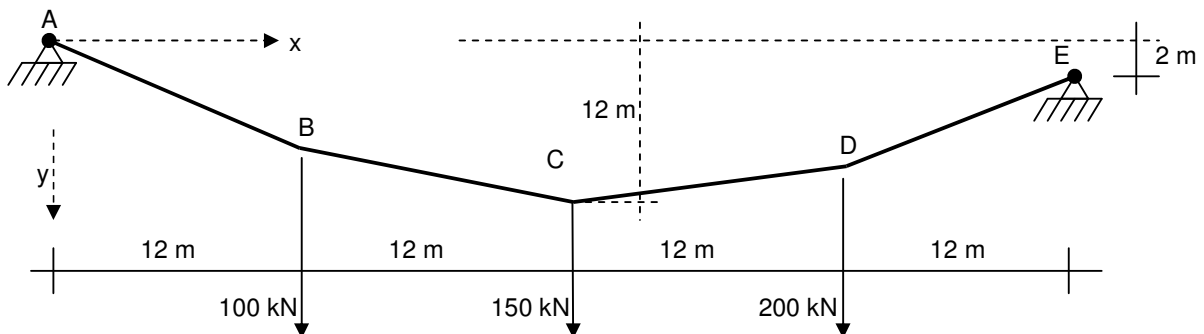
$$L_{BC} = \sqrt{10^2 + 1.563^2} = 10.121 \text{ m}$$

$$L_{CD} = \sqrt{12^2 + 6.563^2} = 13.677 \text{ m}$$

$$\text{Hence } L = L_{AB} + L_{BC} + L_{CD} = 33.232 \text{ m}$$

Example 2: Analyse the cable shown below and determine: -

- (a) The support reaction components at A and E.
- (b) The maximum tension induced in the cable.
- (c) The ordinate of the points B and D.
- (d) The required cable length



Solution:

- (a) Isolate the whole cable; then isolate the portion C – D – E

$$V_A = 213.64 \text{ kN}; H_A = 327.27 \text{ kN}; V_E = 236.36 \text{ kN}; \text{ and } H_E = 327.27 \text{ kN.}$$

- (b) $T_{\max} = T_{DE} = \sqrt{(H_E^2 + V_E^2)} = \sqrt{(327.27^2 + 236.36^2)} = 403.70 \text{ kN}$

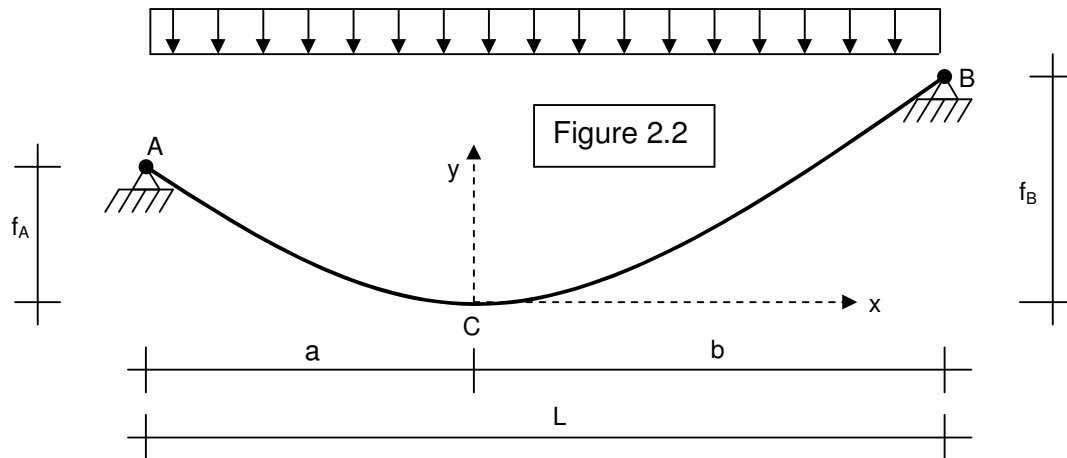
- (c) $y_B = 7.834 \text{ m}; y_C = 12 \text{ m}; \text{ and } y_D = 10.667 \text{ m}$ [All measured from level of A]

- (d) $L_{AB} = 14.331 \text{ m}; L_{BC} = 12.703 \text{ m}; L_{CD} = 12.074 \text{ m}; \text{ and } L_{DE} = 14.803 \text{ m}$

$$\text{Hence } L = 53.918 \text{ m}$$

2.3 CABLE SUBJECTED TO UNIFORMLY DISTRIBUTED LOAD

Figure 2.2 below shows the general model, dimensions and notations, for a cable subjected to a uniformly distributed load ω .



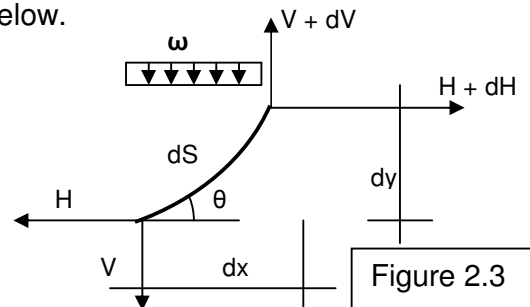
The reference x-y coordinate system is chosen to have its origin at the lowest point C so that the cable profile conforms to a **perfect parabola**.

2.3.1 Cable Profile

Consider a free body diagram of an element (i.e. small portion) of the cable (which is magnified for clarity only) as shown in figure 2.3 below.

The following assumptions are made: -

1. The element is small enough to be straight.
2. The cable self weight is negligible.



For this free body to be in static equilibrium: -

$$\text{➤ } \Sigma F_x = -H + H + dH = 0$$

- Thus $dH = 0$, which means that **H remains constant**

$$\text{➤ } \Sigma F_y = -V + V + dV - \omega dx = 0; \text{ Thus } dV = \omega dx$$

$$\omega = dV/dx \quad \text{----- Eqn (i)}$$

- $$\text{➤ In addition the principle of compatibility requires that } \tan \theta = V/H = dy/dx$$

$$\text{hence we can express; - } V = H (dy/dx) \quad \text{----- Eqn (ii)}$$

- $$\text{➤ Combining equations (i) and (ii), we obtain } \omega = d/dx \{H dy/dx\}$$

$$\text{Hence } \omega/H = d^2y/dx^2 \quad \text{----- Eqn (iii)}$$

Note that in equation (iii), both ω and H are constants and the equation (iii) is therefore a standard 2nd order differential equation whose solution may be obtained by integrating twice and applying the existing boundary conditions.

Thus by integrating equation (iii) twice we obtain:

➤ $dy/dx = (\omega/H)x + C_1$, and

➤ $y = (\omega/2H)x^2 + C_1x + C_2$

Referring to the $x - y$ coordinate system shown in figure 2.2 it can be seen that the existing boundary conditions are: -

✓ $y = 0$ when $x = 0$, so that $C_2 = 0$, and

✓ $dy / dx = 0$ when $x = 0$, so that $C_1 = 0$

Equation 1 is used to
evaluate **a, b and H**

Hence the cable profile is given by the equation:

$y = \omega x^2 / 2H$ ----- **Eqn 1**

Equation 1 is the equation of a perfect parabola: $y = k x^2$; where $k = (\omega/2H)$.

2.3.2 Maximum and Minimum Tension

Considering the free body diagram shown in figure 2.3 it was noted in eqn (i) that

$\omega = dV/dx$, which can be re-arranged thus: -

$dV = \omega dx$ -----Eqn (iv)

Equation (iv) can be solved by integrating once, and applying the existing boundary condition, thus: $V = \omega x + C_3$

Referring to the cable profile shown in figure 2.2, it can be seen that at the point C the cable is horizontal and hence the tensile force in the cable is due to the horizontal component H only. Thus the existing boundary condition is that:

$V = 0$ when $x = 0$, so that $C_3 = 0$

Equation 2 is used to
evaluate **V_A and V_B**

Hence we express: **$V = \omega x$** ----- **Eqn 2**

Equation 2 shows that **V** varies linearly with the distance **x** as measured from point C.

The tension at any point on the cable is given by: $T = \sqrt{(H^2 + V^2)}$

Since H is constant, we can express the tension at any point on the cable as:

$T = \sqrt{H^2 + (\omega x)^2}$ ----- **Eqn 3**

Equation 3 shows that maximum tension occurs when x attains its largest value.

Referring to the general model shown in figure 2.2 where **$a < b$**

$T_{\max} = T_B = \sqrt{H^2 + (\omega b)^2}$ ----- **Eqn 4**

Similarly, minimum tension occurs when $x = 0$: **$T_{\min} = T_C = H$** ----- **Eqn 4***

If **$a = b$** , then **$T_A = T_B = T_{\max}$** .

2.3.3 Support Reaction Components

The horizontal component $H = H_A = H_B$ is obtained by applying Eqn 1 to the given control points. The vertical reaction components V_A and V_B are obtained from Eqn 2. This approach is based on **second principles**.

Alternatively, once **a** and **b** are known, the reaction components can be obtained by applying the eqns of eqb on a fbd of the whole cable. This approach is based on **first principles**. It is recommended that second principles take priority where applicable.

2.3.4 Cable Length: Exact Formulae

Consider, once again, the free body diagram shown in figure 2.3. The **elemental length** is given as **ds**, which may be expressed as: -

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + (dy/dx)^2} dx$$

The cable length is then given by the integral of **ds** over the whole span thus:

$$L = \int_L ds = \int_L \sqrt{1 + (dy/dx)^2} dx \quad \text{-----} \quad \text{Eqn (v)}$$

Since $y = (\omega/2H) x^2$, then $dy/dx = (\omega x/H)$, so that

$$ds = \sqrt{1 + (\omega x/H)^2} dx \quad \text{-----} \quad \text{Eqn (vi)}$$

The cable length from point C to point B is calculated using integral calculus as

$$L_{CB} = \int_0^b \sqrt{1 + (\omega x/H)^2} dx \quad \text{-----} \quad \text{Eqn (vii)}$$

In calculus $\int_0^b \sqrt{1 + (kx)^2} dx = (1/2)b \sqrt{1 + (kb)^2} + (1/2k) \sinh^{-1}(kb)$

In our case $k = \omega/H$

Equation (vii) is therefore a standard equation in calculus, whose solution is given as

$$L_{CB} = \frac{1}{2} b \sqrt{1 + (\omega b/H)^2} + (H/2\omega) \sinh^{-1}(\omega b/H) \quad \text{-----} \quad \text{Eqn 5}$$

Similarly

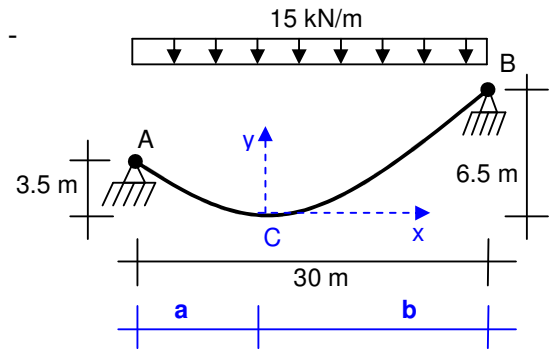
$$L_{CA} = \frac{1}{2} a \sqrt{1 + (\omega a/H)^2} + (H/2\omega) \sinh^{-1}(\omega a/H) \quad \text{-----} \quad \text{Eqn 5}$$

Eqns 5, referred to as the **exact formulae**, are used to calculate the required cable length.

Example 3

Considering the cable shown alongside, determine: -

- The support reaction components.
- The max and min tension induced in the cable.
- The required cable length



Solution

- (a) Let A to C be **a**, and C to B be **b**. Given $f_A = 3.5$ m, $f_B = 6.5$ m, $\omega = 15$ kN/m, and $L = 30$ m;

Using equation 1, $y = \omega x^2 / (2H)$, we have, $f_A = \omega a^2 / (2H)$ and $f_B = \omega b^2 / (2H)$

which may be re-arranged to make H the subject, thus:

$$H = \omega a^2 / (2f_A) \text{ and } H = \omega b^2 / (2f_B)$$

dividing one equation by the other gives: $(a/b)^2 = f_A / f_B$ ----- eqn i

Compatibility requires that: **$a + b = 30$** -----eqn ii

Solving eqns i and ii simultaneously gives

$$a = 12.697 \text{ m, } b = 17.303 \text{ m and } H = 345.46 \text{ kN} = H_A = H_B.$$

The vertical support reaction components are obtained using eqn 3, thus

$$V_A = \omega a = 190.46 \text{ kN; and } V_B = \omega b = 259.55 \text{ kN.}$$

- (b) The max tension occurs at B and is given by $T_{\max} = T_B = \sqrt{H^2 + V_B^2} = 432.10 \text{ kN}$

The min tension is given by $T_{\min} = H = 345.46 \text{ kN}$

- (c) Cable length

$$L_{CA} = (a/2) \sqrt{1 + (\omega a/H)^2} + (H/2\omega) \sinh^{-1} (\omega a/H) = 13.314 \text{ m}$$

$$L_{CB} = (b/2) \sqrt{1 + (\omega b/H)^2} + (H/2\omega) \sinh^{-1} (\omega b/H) = 18.815 \text{ m}$$

$$\text{Hence } L_{ACB} = 13.314 + 18.815 = 32.129 \text{ m}$$

2.3.5 Cable Length: Approximate Formulae

Consider the case of a symmetric cable as shown in figure 2.4.

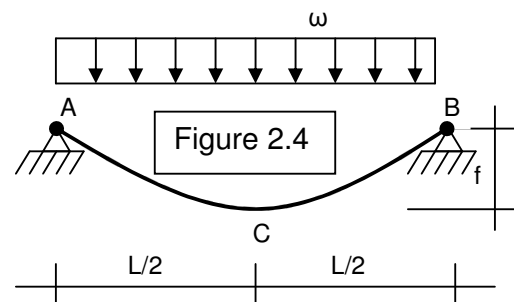
In this case **$a = b = L/2$** , and **$f_A = f_B = f$** , say.

Using equation 1, $y = (\omega/2H)x^2$, we can write

$$f = (\omega/2H) (L/2)^2 = \omega L^2/8H \text{ ----- Eqn (viii)}$$

Re-arranging equation (viii) we obtain

$$H = \omega L^2/8f \text{ ----- Eqn 6}$$



Using equation 3, $T = \sqrt{H^2 + (\omega x)^2}$, we can write

$$T_A = T_B = T_{\max} = \sqrt{H^2 + (\omega L/2)^2} \text{ -----Eqn (ix)}$$

By substituting equation 6 into equation (ix) and simplifying we obtain

$$T_{\max} = T_A = T_B = (\omega L^2/8f) \sqrt{1 + 16(f/L)^2} \text{ ----- Eqn 7}$$

Using Eqn 6 we can write $\omega/2H = 4f/L^2$ ----- Eqn (x)

Substitute Eqn (x) into Eqn 1

$$y = (4f/L^2)x^2 \text{ so that } dy/dx = (8f/L^2) x.$$

The cable length may then be calculated using equation (v), thus

$$L_{CB} = \int_0^{L/2} \sqrt{1 + (dy/dx)^2} dx = \int_0^{L/2} \{1 + (64f^2/L^4) x^2\}^{1/2} dx \text{ ----- Eqn (xi)}$$

Equation (xi) can be solved using the general Binomial Theorem.

The Binomial Theorem is used to expand the above expression into an infinite series, thus

$$\{1 + x\}^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \text{-----}$$

provided n is any number and $|x| < 1$

The binomial expansion converges pretty fast, and only a few terms will be considered.

Using the general binomial expansion, equation (xi) may be written as:

$$\begin{aligned} L_{CB} &= \int_0^{L/2} \left\{ 1 + \frac{32 f^2 x^2}{L^4} - \frac{8(64) f^4 x^4}{L^8} + \dots \right\} dx \\ &= \left[x + \frac{32 f^2 x^3}{3L^4} - \frac{512 f^4 x^5}{5L^8} + \dots \right]_0^{L/2} \\ &= L/2 + (4/3) (f^2 / L) - (16/5) (f^4 / L^3) + \dots \\ &= L/2 \{ 1 + (8/3) (f/L)^2 - (32/5) (f/L)^4 + \dots \} \end{aligned}$$

In practice $f/L \ll 1$, and the cable length can be approximated by:

$$L_{CB} = L_{CA} = (L/2) \{ 1 + (8/3)(f/L)^2 \} \text{ ----- Eqn 8}$$

Eqn 8 is referred to as the **approximate formulae** for cable length.

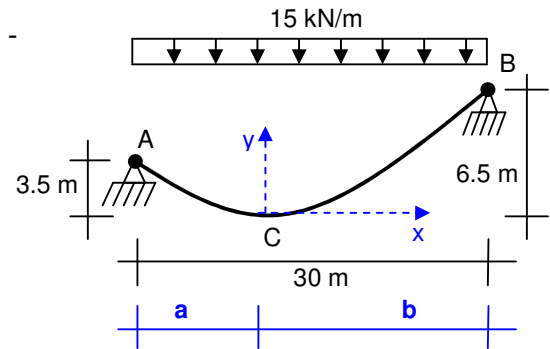
Eqn 5 and Eqn 8 are the same. Both Eqn 5 and Eqn 8 can be used to calculate the cable length for any cable: whether symmetric or non-symmetric.

Example 3

Considering the cable shown alongside, determine: -

- The support reaction components.
- The max and min tension induced in the cable.
- The required cable length, using both the exact formula and the approximate formula
- The expected cable elongation

[Take $E = 200 \text{ kN/mm}^2$ and $A = 576 \text{ mm}^2$]



Solution

- (a) Let A to C be **a**, and C to B be **b**. Given $f_A = 3.5 \text{ m}$, $f_B = 6.5 \text{ m}$, $\omega = 15 \text{ kN/m}$, and $L = 30 \text{ m}$;

Using equation 1, $y = \omega x^2 / (2H)$, we have, $f_A = \omega a^2 / (2H)$ and $f_B = \omega b^2 / (2H)$

which may be re-arranged to make H the subject, thus:

$$H = \omega a^2 / (2f_A) \text{ and } H = \omega b^2 / (2f_B)$$

dividing one equation by the other gives: $(a/b)^2 = f_A / f_B$ ----- eqn i

Compatibility requires that: **$a + b = 30$** -----eqn ii

Solving eqns i and ii simultaneously gives

$$a = 12.697 \text{ m}, b = 17.303 \text{ m} \text{ and } H = 345.46 \text{ kN} = H_A = H_B.$$

The vertical support reaction components are obtained using eqn 3, thus

$$V_A = \omega a = 190.46 \text{ kN}; \text{ and } V_B = \omega b = 259.55 \text{ kN}.$$

- (b) The max tension occurs at B and is given by $T_{\max} = T_B = \sqrt{H^2 + V_B^2} = 432.10 \text{ kN}$

The min tension is given by $T_{\min} = H = 345.46 \text{ kN}$

- (c) (i) Cable length using exact formula:

$$L_{CA} = (a/2) \sqrt{1 + (\omega a/H)^2} + (H/2\omega) \sinh^{-1} (\omega a/H) = 13.314 \text{ m}$$

$$L_{CB} = (b/2) \sqrt{1 + (\omega b/H)^2} + (H/2\omega) \sinh^{-1} (\omega b/H) = 18.815 \text{ m}$$

$$\text{Hence } L_{ACB} = 13.314 + 18.815 = 32.129 \text{ m}$$

- (i) Cable length using approximate formula

$$L_{CA} \approx a \{1 + (8/3) (f_A/L)^2\} \approx 13.158 \text{ m} \text{ and } L_{CB} \approx b \{1 + (8/3) (f_B/L)^2\} \approx 19.469 \text{ m}$$

$$\text{Hence } L_{ACB} \approx 13.158 + 19.469 \approx 32.627 \text{ m}$$

[The percentage error $e = \{(32.627 - 32.129)/32.129\} \times 100\% = 1.55\%$ on the safe side.]

2.4 CABLE ELONGATION

2.4.1 Basic Principle

When a cable is subjected to loading, the material develops stresses which result in strains which make the cable longer than theoretically predicted in articles 2.3.4 and 2.3.5.

Such cable elongation may be predicted as discussed hereunder.

The underlying principle is based on the uni-axial tensile test model illustrated in figure 2.5.

The stress is given by $\sigma = P/A$

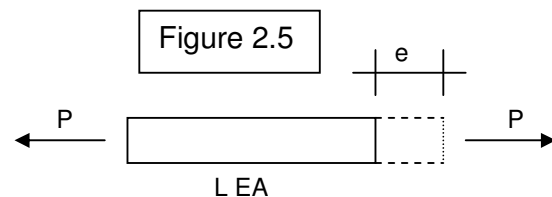
The strain is given by $\epsilon = e/L$

By the principle of constitutivity $\sigma = E\epsilon$

So that $P/A = E (e/L)$, and hence

$$e = PL/EA$$

The quantity EA is called the axial stiffness.



2.4.2 Cable Subjected to Point Loads

Each straight portion of the cable will suffer an extension is given by $e = TL/EA$

The **elongation of the cable subjected to point loads** is therefore calculated by **piece-wise summation**, thus:

$$\Delta s = \sum_{i=1}^n T_i L_i / EA \quad \text{----- Eqn 9}$$

2.4.3 Cable Subjected to UDL

Consider an element of a cable as shown in figure 2.6.

The elemental elongation e is given by $e = (T/EA) ds$

By compatibility $T/H = ds/dx$, so that $T = H ds/dx$,

and the elemental elongation can be written as:

$$e = (H/EA) (ds/dx) ds \quad \text{----- eqn (xii).}$$

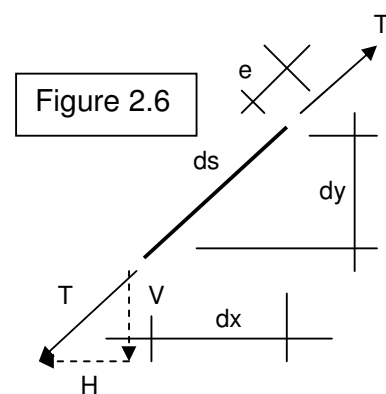
Equation (xii) may be re-written as

$$e = (H/EA) (ds/dx) (ds/dx) dx = (H/EA) (ds/dx)^2 dx \quad \text{----- eqn (xiii)}$$

By geometry $(ds)^2 = (dx)^2 + (dy)^2$, so that $(ds/dx)^2 = 1 + (dy/dx)^2$.

Equation (xii) therefore becomes

$$e = (H/EA) \{1 + (dy/dx)^2\} dx \quad \text{----- Eqn 10}$$



Since $y = (\omega/2H)x^2$ so that $(dy/dx) = (\omega/H)x$, the total elongation Δs is obtained by integrating equation 10 over the whole span, thus

$$\Delta s = (H/EA) \int_L \{1 + (\omega/H)^2 x^2\} dx \quad \text{----- Eqn 11}$$

Considering the **general cable model** shown in figure 2.2, we can write

$$\Delta s = (H/EA) \int_0^a \{1 + (\omega/H)^2 x^2\} dx + (H/EA) \int_0^b \{1 + (\omega/H)^2 x^2\} dx \text{ which gives:}$$

$$\Delta s = (Ha/EA)\{1 + \frac{1}{3}(\omega a/H)^2\} + (Hb/EA)\{1 + \frac{1}{3}(\omega b/H)^2\} \quad \text{----- Eqn 12}$$

Considering the **symmetric cable model** shown in figure 2.4, since $a = b = L/2$, we obtain

$$\Delta s = (HL/EA)\{1 + (1/12)(\omega L/H)^2\} \quad \text{----- Eqn 13}$$

Alternatively: -

For a symmetrical cable, $y = (4f/L^2)x^2$, so that $dy/dx = (8f/L^2)x$, and equation 11 gives

$$\Delta s = 2 (H/EA) \int_0^{L/2} \{1 + (64f^2 / L^4) x^2\} dx \quad \text{----- Eqn 14}$$

Equation 14, upon integration, gives

$$\Delta s = (HL/EA)\{1 + (16/3)(f/L)^2\} \quad \text{----- Eqn 15}$$

Eqns 12, 13 and 15 are the same. Any of these Eqns can be used to calculate the cable elongation: whether the cable be symmetric or non-symmetric.

2.4.4 Cable Elongation due to Temperature Change

A rise in temperature by, say, T °C will result in cable elongation given by $\Delta s = \alpha LT$ where α is the **coefficient of linear thermal expansion** of the material, and L is the **cable length**. Such an elongation results in an **increase in the cable sag** and a **reduction of the tensile stresses** set up in the cable. The cable is said to become **lax**.

On the other hand, a **drop in temperature** makes the cable contract thereby shortening. Such shortening results in a **decrease in the cable sag** and an **increase in the tensile stresses** set up in the cable. In that case, the cable is said to become **taut**.

Solution to Example 3 (d)

$H = 345.46$ kN, $a = 12.697$ m, $b = 17.303$ m, $\omega = 15$ kN/m, $f_A = 3.5$ m, $f_B = 6.5$ m, and $EA = 115.2 \times 10^3$ kN

$$\text{ALT 1: Eqn 12: } \Delta s = (Ha/EA) \{1 + \frac{1}{3}(\omega a/H)^2\} + (Hb/EA) \{1 + \frac{1}{3}(\omega b/H)^2\} = 103.6 \text{ mm}$$

$$\text{ALT 2: Eqn 13: } \Delta s = (Ha/EA) \{1 + (1/12)(\omega 2a/H)^2\} + (Hb/EA) \{1 + (1/12)(\omega 2b/H)^2\} = 103.6 \text{ mm}$$

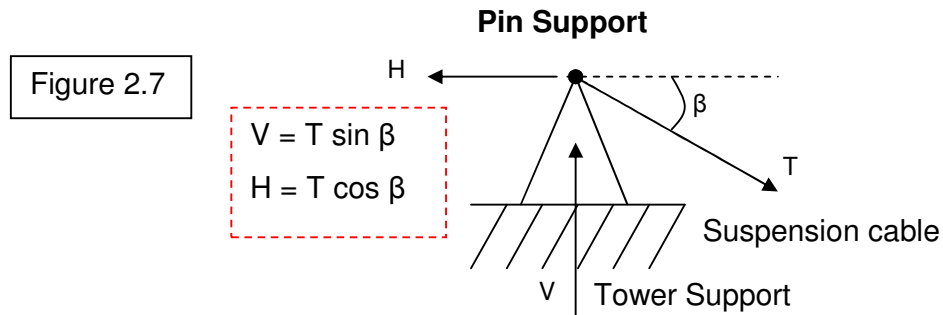
$$\text{ALT 3: Eqn 15: } \Delta s = (Ha/EA) \{1 + (16/3)(f_A/2a)^2\} + (Hb/EA) \{1 + (16/3)(f_B/2b)^2\} = 103.6 \text{ mm}$$

2.5 SUPPORT ARRANGEMENT

There are three ways in which a suspension cable may be supported.

2.5.1 Pin Supports

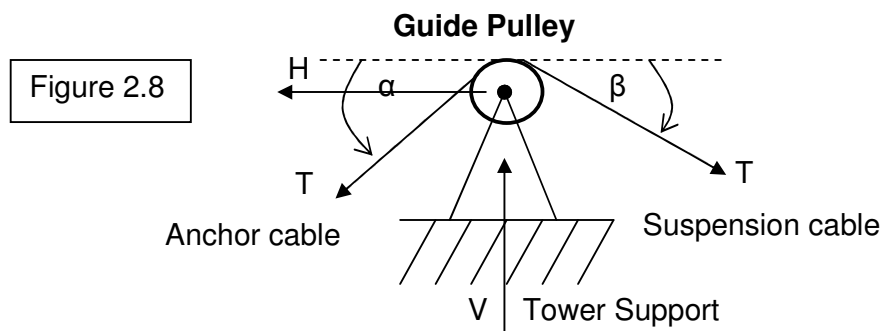
The towers may be designed to act as pin supports to the cable. In that case the towers transmit the vertical and horizontal components of the tension in the cable by compression and bending respectively.



2.5.2 Guide Pulley

The suspension cable is passed over a frictionless guide pulley mounted on the tower and the anchor cable is pinned to concrete mass which is sufficient to resist the horizontal component by ground friction. The anchor cable may be loaded, and therefore curved, or may be straight. But in general, if α and β are the inclinations of the anchor cable and suspension cable respectively as they exit the guide pulley at the tower, then:

1. The tension in the anchor cable is equal in magnitude to that in the suspension cable.
2. The vertical reaction on the tower is given by: $V = T(\sin \alpha + \sin \beta)$
3. And the net horizontal reaction on the tower is given by: $H = T(\cos \beta - \cos \alpha)$



2.5.3 Roller Support

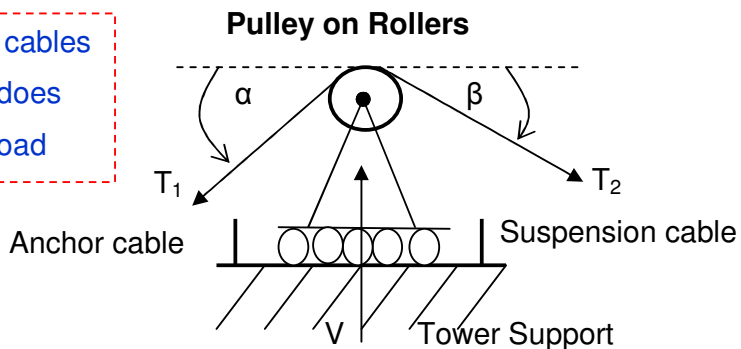
The suspension cable is passed over a saddle mounted on rollers resting on the tower and the anchor cable is pinned to concrete mass which is sufficient to resist the horizontal component by ground friction. The anchor cable may be loaded, and therefore curved, or

may be straight. But in general, if α and β are the inclinations of the anchor cable and suspension cable respectively as they exit the guide pulley at the tower, then:

1. The horizontal component H in the anchor cable is equal in magnitude and opposite in sense to that in the suspension cable but the vertical components will vary.
2. The vertical reaction on the tower is given by: $V = T_1 \sin \alpha + T_2 \sin \beta$
3. And the net horizontal reaction on the tower is given by: $T_1 \cos \alpha = T_2 \cos \beta$

The H components in the cables cancel out and the tower does not suffer any horizontal load

Figure 2.9



2.6 SUMMARY

1. Cable subjected to point loads

- ✓ Application of static equilibrium to solve for reactions, ordinates and tension.
- ✓ Application of Pythagoras theorem to calculate cable length
- ✓ Application of Hooke's law to calculate cable elongation

2. Cable subjected to u.d.l

- ✓ $y = (\omega/2H)x^2$, $H = \text{constant}$, $V = \omega x$, $T = \sqrt{H^2 + (\omega x)^2}$, $T_{\max}$ occurs when x is largest, $T_{\min} = H$
- ✓ for a symmetric cable: $y = 4fx^2/L^2$; $H = \omega L^2/8f$, $T_{\max} = (\omega L^2/8f) \sqrt{1 + 16(f/L)^2}$

3. Cable length

- ✓ $L_{CA} = \frac{1}{2} a \sqrt{1 + (\omega a/H)^2} + (H/2\omega) \sinh^{-1} (\omega a/H)$
- ✓ $L_{CB} = \frac{1}{2} b \sqrt{1 + (\omega b/H)^2} + (H/2\omega) \sinh^{-1} (\omega b/H)$
- ✓ For a symmetrical cable: $L_{CA} = L_{CB} \approx \frac{1}{2} L \{1 + (8/3)(f/L)^2\}$

4. Cable elongation

- ✓ $\Delta s = (Ha/EA) \{1 + \frac{1}{3} (\omega a/H)^2\} + (Hb/EA) \{1 + \frac{1}{3} (\omega b/H)^2\}$
- ✓ For a symmetric cable: $\Delta s = (HL/EA) \{1 + (1/12)(\omega L/H)^2\}$
- ✓ Alternatively: $\Delta s = (HL/EA) \{1 + (16/3)(f/L)^2\}$

3.0 THE ARCH

3.1 INTRODUCTION

The shape of an **ideal arch** is the inverse of the shape of a cable subjected to the same loads. Such an ideal arch would be a **purely compression structure**. The arch is not form-active. In practice, the ideal arch profile is hard to achieve because of the required span to height ratio. **Most arches are therefore non-ideal**, and will carry the applied loads by developing not only **axial compression**, but also **shear force**, and **bending moment**.

The **masonry arch** is among the earliest forms of construction. Today, arches may be constructed using masonry, reinforced concrete, steel, laminated timber, or any combination thereof.

Arches find **applications** in many areas including: **Industrial low-rise sheds**, Highway **bridges**, via-ducts, foot bridges, **Large-span roof systems** (stadium, lecture theatre), **Architectural works** in churches, mosques, libraries, shopping malls, **civil block**, etc

Arches may take various **geometric shapes** as illustrated in figure 3.1(a). The parabolic shape very efficient for uniformly distributed loading. The parabolic arch transmits the uniformly distributed load by developing axial stresses only; no shear and no moments.

Arches may be constructed with **varying degrees of static indeterminacy**, as illustrated in figure 3.1(b) below. In FCE 201 we shall deal with statically determinate arches.

Fig. 3.1(a)

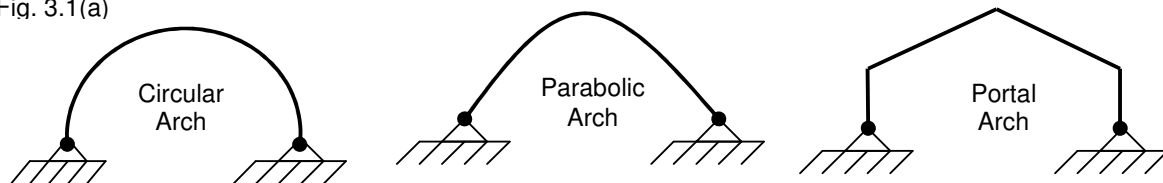
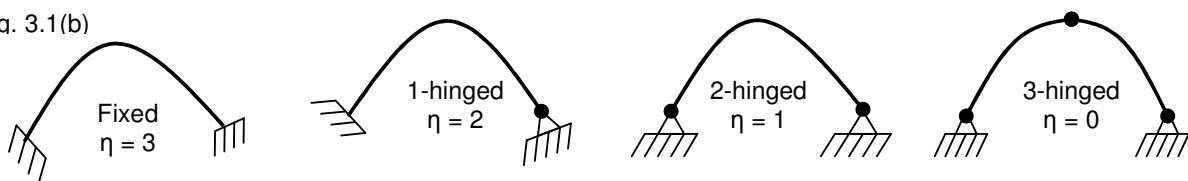


Fig. 3.1(b)



3.2 STRUCTURAL ANALYSIS

Structural analysis of arches involves determining: -

1. The arch **profile**.
2. The **support reaction components**.
3. The **axial force N , shear force V , and bending moment M** induced at any given cross-section.
4. The **displacements** of the salient points on the arch profile.

Once the arch profile is determined, items 2 and 3 of the analysis can be achieved by application of the principle of **static equilibrium**.

The **general procedure** for the analysis of arches may be outlined as follows:-

1. Set up an appropriate x-y coordinate system; and determine the arch profile. Hence determine, at any given cross-section, the coordinates and the slope.
2. Evaluate the support reaction components.
3. For any given cross-section: -
 - ✓ Isolate an appropriate free body diagram,
 - ✓ Apply equilibrium to generate three equations in N , V , and M .
 - ✓ Solve simultaneously.

The above procedure is illustrated in Examples 1, 2 and 3 below.

The axial force **N** , shear force **V** , and bending moment **M** are **continuous (at least piece-wise) mathematical functions** over the span of the arch. These functions may be shown as **graphical variations** of the load effect under consideration, from cross-section to cross-section along the arch or along the span. Such graphical representations are referred to as **diagrams**. The axial force diagram **AFD**, shear force diagram **SFD**, bending moment diagram **BMD**, and the **D-shape** are dealt with in Chapter 5.

Arches are predominantly compression structures. In order to avoid buckling, the member sizes are usually bulky. Consequently the arch suffers very small axial deformations. However, in certain arch types, the displacements due to flexure may be significant and these shall be dealt with in FCE 202.

Example 1

Determine **N**, **V** and **M** induced at B and D on the **pitched portal arch** shown below.

Solution

Support reaction components

$$\Sigma F_x = H_A - H_E + 216 = 0$$

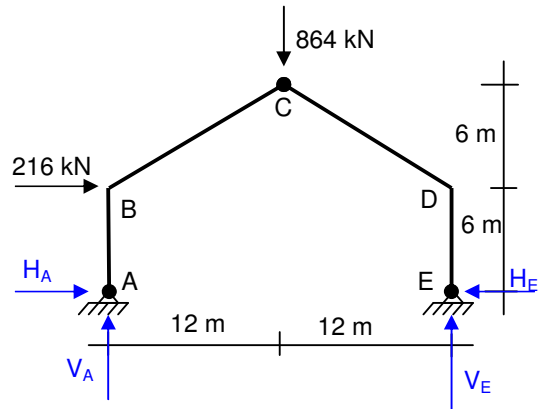
$$\Sigma F_y = V_A + V_E - 864 = 0$$

$$\Sigma M_A = 216(6) + 864(12) - V_E(24) = 0$$

$$\Sigma M_{C(R)} = H_E(12) - V_E(12) = 0$$

$$\text{Solving: } V_A = 378 \text{ kN} \quad V_E = 486 \text{ kN}$$

$$H_A = 270 \text{ kN} \quad H_E = 486 \text{ kN}$$



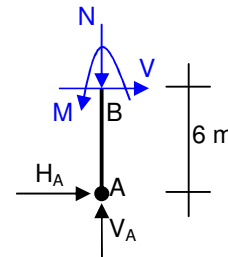
Note: The values of **N** and **V** will change as you arrive and pass the point B (or Point D). We therefore think of **just below** and **just above** the point B (or D), where the word **just** means zero distance from the point.

Just below point B

$$\Sigma F_N = N - V_A = 0 \Rightarrow N = V_A = 378 \text{ kN}$$

$$\Sigma F_V = V + H_A = 0 \Rightarrow V = -H_A = -270 \text{ kN}$$

$$\Sigma M_X = H_A(6) + M = 0 \Rightarrow M = -H_A(6) = -1620 \text{ kNm}$$



Just above point B ($\theta = \tan^{-1}(6/12) = 26.57^\circ$)

$$\Sigma F_N = N - V_A \sin \theta - H_A \cos \theta - 216 \cos \theta = 0$$

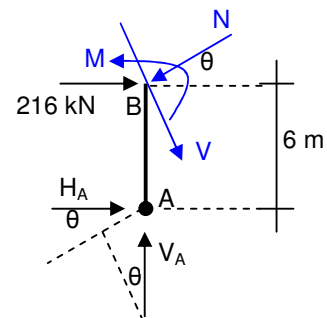
$$\Rightarrow N = V_A \sin \theta + H_A \cos \theta + 216 \cos \theta = 603.7 \text{ kN}$$

$$\Sigma F_V = V - V_A \cos \theta + H_A \sin \theta + 216 \sin \theta = 0$$

$$\Rightarrow V = V_A \cos \theta - H_A \sin \theta - 216 \sin \theta = 120.7 \text{ kN}$$

$$\Sigma M_X = H_A(6) + M = 0$$

$$\Rightarrow M = -H_A(6) = -1620 \text{ kNm}$$

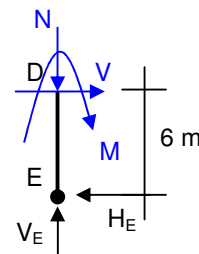


Just below point D

$$\Sigma F_N = N - V_E = 0 \Rightarrow N = V_E = 486 \text{ kN}$$

$$\Sigma F_V = V - H_E = 0 \Rightarrow V = H_E = 486 \text{ kN}$$

$$\Sigma M_X = H_E(6) + M = 0 \Rightarrow M = -H_E(6) = -2916 \text{ kNm}$$



Just above point D ($\theta = \tan^{-1} (6/12) = 26.57^\circ$)

$$\Sigma F_N = N - V_E \sin \theta - H_E \cos \theta = 0$$

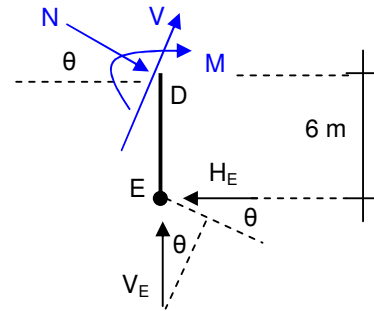
$$\Rightarrow N = V_A \sin \theta + H_A \cos \theta = 652.1 \text{ kN}$$

$$\Sigma F_V = V + V_E \cos \theta - H_E \sin \theta = 0$$

$$\Rightarrow V = -V_E \cos \theta + H_A \sin \theta = -217.3 \text{ kN}$$

$$\Sigma M_X = H_E(6) + M = 0$$

$$\Rightarrow M = -H_E(6) = -2916 \text{ kNm}$$



Note: The sign convention for N , V and M , and their representation on free body diagrams, is explained in Chapter 5.

Example 2

Determine N , V and M induced at point D in the **semi-circular arch** shown below

Solution

Support reaction components

$$\Sigma F_x = H_A - H_B = 0$$

$$\Sigma F_y = V_A + V_B - 36(24) = 0$$

$$\Sigma M_A = 36(24)(12) - V_B(12) = 0$$

$$\Sigma M_{C(R)} = (36)(12)(6) + H_B(12) - V_B(12) = 0$$

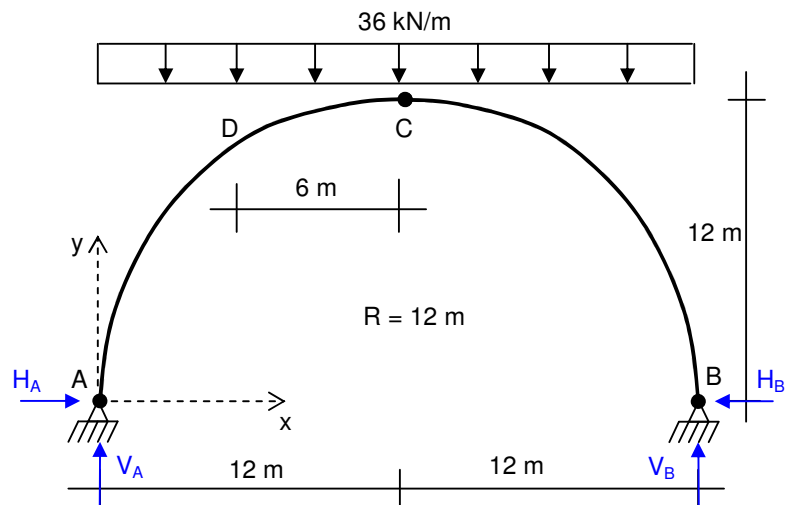
Solving:

$$V_A = 432 \text{ kN}$$

$$H_A = 216 \text{ kN}$$

$$V_B = 432 \text{ kN}$$

$$H_B = 216 \text{ kN}$$



Arch profile (sing the x-y system shown)

$$y = \sqrt{(2Rx - x^2)}$$

$$dy/dx = (R - x)/\sqrt{(2Rx - x^2)}$$

$$\Rightarrow \theta = \tan^{-1} \{(R - x)/\sqrt{(2Rx - x^2)}\}$$

N, V and M at D

$$y = \sqrt{(2Rx - x^2)} = 10.392 \text{ m};$$

$$\theta = \tan^{-1} \{(R - x)/\sqrt{(2Rx - x^2)}\} = 30^\circ$$

$$\Sigma F_N = N - V_A \sin \theta - H_A \cos \theta + 36(6) \sin \theta = 0$$

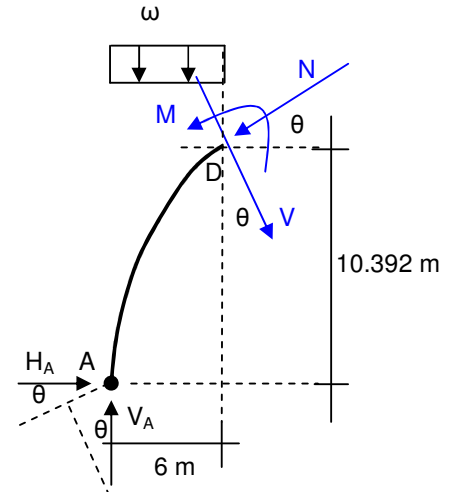
$$\Rightarrow N = V_A \sin \theta + H_A \cos \theta - 36(6) \sin \theta = 295.1 \text{ kN}$$

$$\Sigma F_V = V - V_A \cos \theta + H_A \sin \theta + 36(6) \cos \theta = 0$$

$$\Rightarrow V = V_A \cos \theta - H_A \sin \theta - 36(6) \cos \theta = 79.1 \text{ kN}$$

$$\Sigma M_X = V_A(6) - H_A(10.392) - 36(6)(3) - M = 0$$

$$\Rightarrow M = V_A(6) - H_A(10.392) - 36(6)(3) = -300.7 \text{ kN}$$



Example 3

Determine **N**, **V**, and **M** induced at A, B, C, D, E, F, and G in the **parabolic arch** shown below.

Solution

Using the x-y coordinate system shown, the arch profile is given by:

$$y = kx^2$$

$$\text{At A, } y = 12 \text{ m and } x = 24 \text{ m}$$

$$\text{Thus } 12 = k(24)^2 \Rightarrow k = 1/48$$

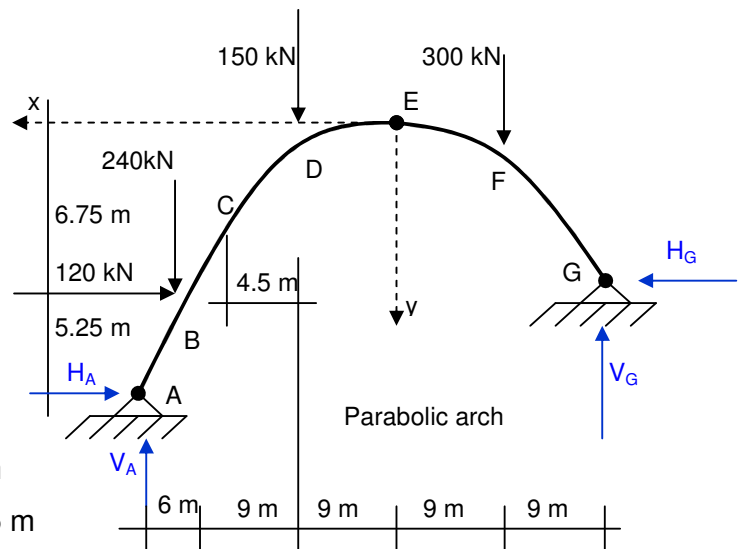
$$\text{Hence } y = x^2/48 \text{ and } dy/dx = x/24$$

$$\text{At D, } x = 9 \text{ m} \Rightarrow y_D = 1.6875 \text{ m}$$

$$\text{Also at F, } x = 9 \text{ m} \Rightarrow y_F = 1.6875 \text{ m}$$

$$\text{At C, } x = 13.5 \text{ m} \Rightarrow y_C = 3.7969 \text{ m}$$

$$\text{At B, } x = 18 \text{ m and } y_B = 6.75 \text{ m, same as at G.}$$



Isolate the whole arch as a free body diagram

$$\Sigma F_x = H_A + 120 - H_G = 0$$

$$\Sigma F_y = V_A + V_G - 690 = 0$$

$$\Sigma M_A = 120 \times 5.25 + 240 \times 6 + 150 \times 15 + 300 \times 33 - V_G \times 42 - H_G \times 5.25 = 0$$

$$\Sigma M_{C(R)} = 300 \times 9 + H_G \times 6.75 - V_G \times 18 = 0$$

Solving: **$V_A = 398.6 \text{ kN}$, $H_A = 257.1 \text{ kN}$**

$$\mathbf{V_G = 291.4 \text{ kN and } H_G = 377.1 \text{ kN}}$$

At A: Isolate a fbd of the arch portion *just above point A*

$$\tan \theta_A = dy/dx|_A = 24/24 \Rightarrow \theta_A = 45^\circ$$

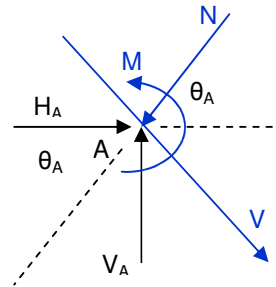
$$\sum F_N = N - V_A \sin \theta_A - H_A \cos \theta_A = 0$$

$$\Rightarrow \mathbf{N = 463.6 \text{ kN}}$$

$$\sum F_V = V - V_A \cos \theta_A + H_A \sin \theta_A = 0$$

$$\Rightarrow \mathbf{V = 100.1 \text{ kN}}$$

$$\sum M_A = M = 0$$



Just to the left of B

Isolate a fbd of the arch portion A to B (*the loads at B will not feature*)

$$\tan \theta_B = dy/dx|_B = 18/24 \Rightarrow \theta_B = 36.87^\circ$$

$$\sum F_N = N - V_A \sin \theta_B - H_A \cos \theta_B = 0$$

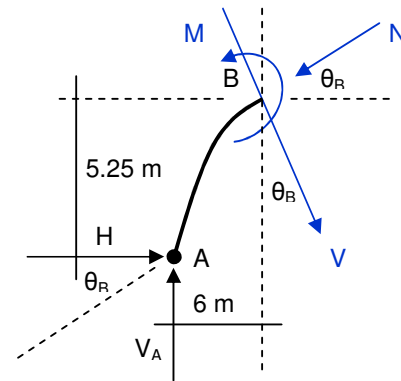
$$\Rightarrow \mathbf{N = 444.4 \text{ kN}}$$

$$\sum F_V = V - V_A \cos \theta_B + H_A \sin \theta_B = 0$$

$$\Rightarrow \mathbf{V = 164.6 \text{ kN}}$$

$$\sum M_B = V_A \times 6 - H_A \times 5.25 - M = 0$$

$$\Rightarrow \mathbf{M = 1041.8 \text{ kNm}}$$



Just to the right of B

Isolate a fbd of the arch portion A to B (*the loads at B will feature*)

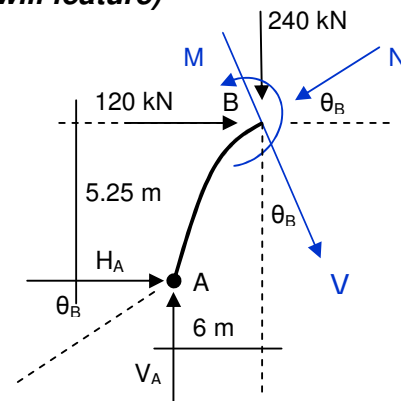
$$\tan \theta_B = dy/dx|_B = 18/24 \Rightarrow \theta_B = 36.87^\circ$$

$$\sum F_N = N - V_A \sin \theta_B - H_A \cos \theta_B - 120 \cos \theta_B + 240 \sin \theta_B = 0 \Rightarrow \mathbf{N = 396.8 \text{ kN}}$$

$$\sum F_V = V - V_A \cos \theta_B + H_A \sin \theta_B + 120 \sin \theta_B + 240 \cos \theta_B = 0 \Rightarrow \mathbf{V = -99.4 \text{ kN}}$$

$$\sum M_D = V_A \times 6 - H_A \times 5.25 - M = 0$$

$$\Rightarrow \mathbf{M = 1041.8 \text{ kNm}}$$



At C: Isolate a fbd of the arch portion A to C

$$\tan \theta_C = dy/dx|_C = 13.5/24 \Rightarrow \theta_C = 29.36^\circ$$

Using the N-V axes system:

$$\sum F_N = N - V_A \sin \theta_C - H_A \cos \theta_C - 120 \cos \theta_C$$

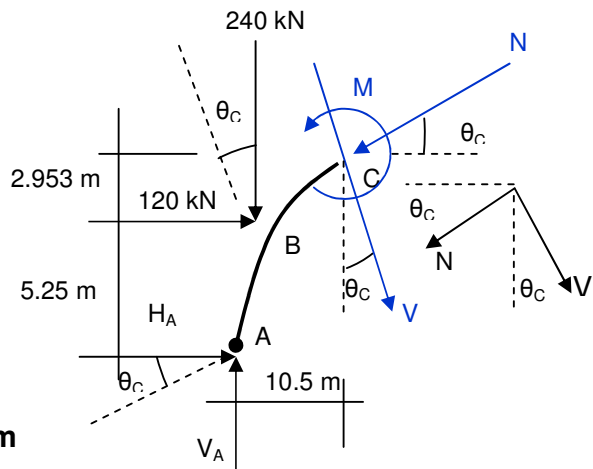
$$+ 240 \sin \theta_C = 0 \Rightarrow \mathbf{N = 406.4 \text{ kN}}$$

$$\sum F_V = V - V_A \cos \theta_C + H_A \sin \theta_C + 120 \sin \theta_C$$

$$+ 240 \cos \theta_C = 0 \Rightarrow \mathbf{V = -46.7 \text{ kN}}$$

$$\sum M_C = V_A \times 10.5 - H_A \times 8.203 - 120 \times 2.953$$

$$- 240 \times 4.5 - M = 0 \Rightarrow \mathbf{M = 641.9 \text{ kNm}}$$



Just to the left of D (the 150 kN will not feature)

Isolate a fbd of the arch portion A to D.

$$\tan \theta_D = dy/dx|_D = 9/24 \Rightarrow \theta_D = 20.56^\circ$$

$$\sum F_N = N - V_A \sin \theta_D - H_A \cos \theta_D - 120 \cos \theta_D$$

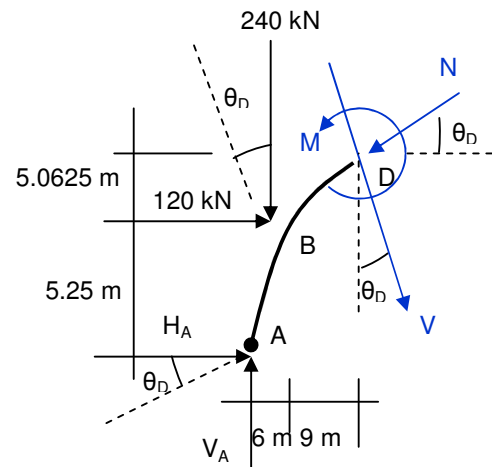
$$+ 240 \sin \theta_D = 0 \Rightarrow \mathbf{N = 408.8 \text{ kN}}$$

$$\sum F_V = V - V_A \cos \theta_D + H_A \sin \theta_D + 120 \sin \theta_D$$

$$+ 240 \cos \theta_D = 0 \Rightarrow \mathbf{V = 16.1 \text{ kN}}$$

$$\sum M_D = V_A \times 15 - H_A \times 10.3125 - 120 \times 5.0625$$

$$- 240 \times 9 - M = 0 \Rightarrow \mathbf{M = 560.2 \text{ kNm}}$$



Just to the right of D (the 150 kN will feature).

Isolate a fbd of the arch portion A to D

$$\tan \theta_D = dy/dx|_D = 9/24 \Rightarrow \theta_D = 20.56^\circ$$

$$\sum F_N = N - V_A \sin \theta_D - H_A \cos \theta_D - 120 \cos \theta_D$$

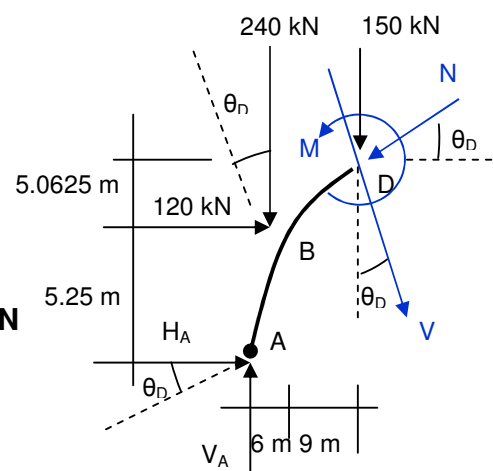
$$+ 240 \sin \theta_D + 150 \sin \theta_D = 0 \Rightarrow \mathbf{N = 356.1 \text{ kN}}$$

$$\sum F_V = V - V_A \cos \theta_D + H_A \sin \theta_D + 120 \sin \theta_D$$

$$+ 240 \cos \theta_D + 150 \cos \theta_D = 0 \Rightarrow \mathbf{V = -124.4 \text{ kN}}$$

$$\sum M_D = V_A \times 15 - H_A \times 10.3125 - 120 \times 5.0625$$

$$- 240 \times 9 - M = 0 \Rightarrow \mathbf{M = 560.2 \text{ kNm}}$$



At E

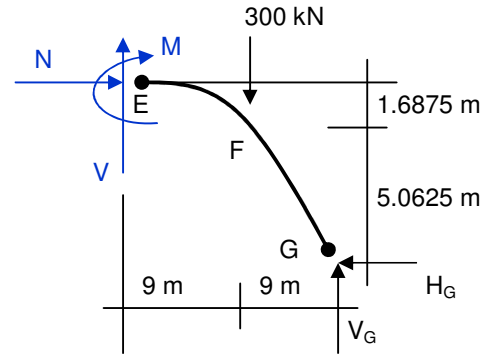
Isolate a fbd of the arch portion E to G

The slope $\theta_E = 0$

$$\sum F_N = N - H_G = 0 \Rightarrow \mathbf{N = 377.1 \text{ kN}}$$

$$\sum F_V = V - 300 + V_G = 0 \Rightarrow \mathbf{V = 8.6 \text{ kN}}$$

$$\begin{aligned} \sum M_E &= M + 300 \times 9 + H_G \times 6.75 - V_G \times 18 = 0 \\ \Rightarrow \mathbf{M} &= \mathbf{0} \end{aligned}$$



Just to the left of F (the 300 kN will feature)

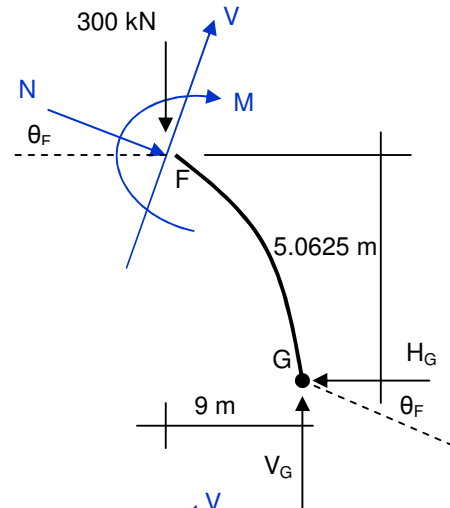
Isolate a fbd of the arch portion G to F

$$\tan \theta_F = dy/dx|_F = 9/24 \Rightarrow \theta_F = 20.56^\circ$$

$$\begin{aligned} \sum F_N &= N - V_G \sin \theta_F - H_G \cos \theta_F + 300 \sin \theta_F = 0 \\ \Rightarrow \mathbf{N} &= \mathbf{350.0 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum F_V &= V + V_G \cos \theta_F - H_G \sin \theta_F - 300 \cos \theta_F = 0 \\ \Rightarrow \mathbf{V} &= \mathbf{140.5 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum M_F &= -V_G \times 9 + H_G \times 5.0625 + M = 0 \\ \Rightarrow \mathbf{M} &= \mathbf{713.5 \text{ kNm}} \end{aligned}$$



Just to the right of F (the 300 kN will not feature)

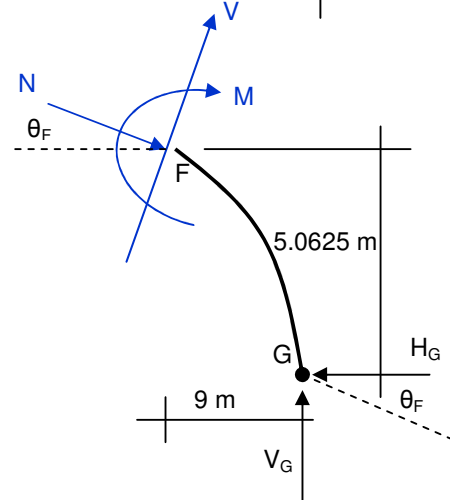
Isolate a fbd of the arch portion G to F

$$\tan \theta_F = dy/dx|_F = 9/24 \Rightarrow \theta_F = 20.56^\circ$$

$$\begin{aligned} \sum F_N &= N - V_G \sin \theta_F - H_G \cos \theta_F = 0 \\ \Rightarrow \mathbf{N} &= \mathbf{455.4 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum F_V &= V + V_G \cos \theta_F - H_G \sin \theta_F = 0 \\ \Rightarrow \mathbf{V} &= \mathbf{-140.4 \text{ kN}} \end{aligned}$$

$$\begin{aligned} \sum M_F &= -V_G \times 9 + H_G \times 5.0625 + M = 0 \\ \Rightarrow \mathbf{M} &= \mathbf{713.5 \text{ kNm}} \end{aligned}$$



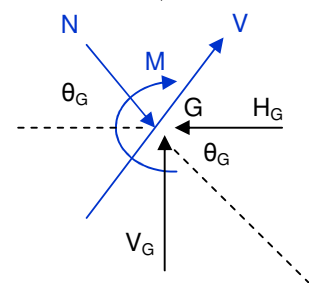
At G: Isolate a fbd of the arch portion just above point G

$$\tan \theta_G = dy/dx|_G = 18/24 \Rightarrow \theta_G = 36.87^\circ$$

$$\sum F_N = N - V_G \sin \theta_G - H_G \cos \theta_G = 0 \Rightarrow \mathbf{N = 476.5 \text{ kN}}$$

$$\sum F_V = V + V_G \cos \theta_G - H_G \sin \theta_G = 0 \Rightarrow \mathbf{V = -6.9 \text{ kN}}$$

$$\sum M_G = \mathbf{M = 0}$$

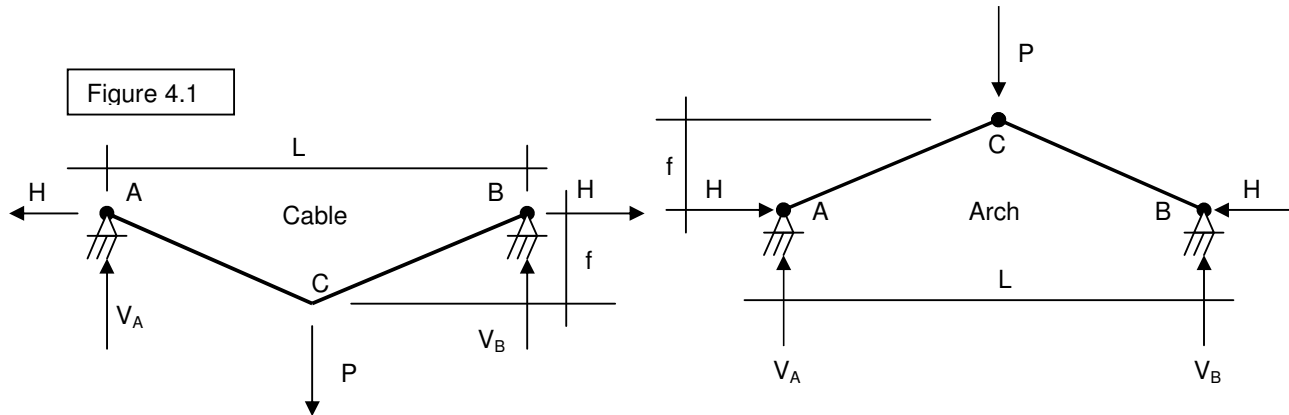


4 THE TRUSS

4.1 INTRODUCTION

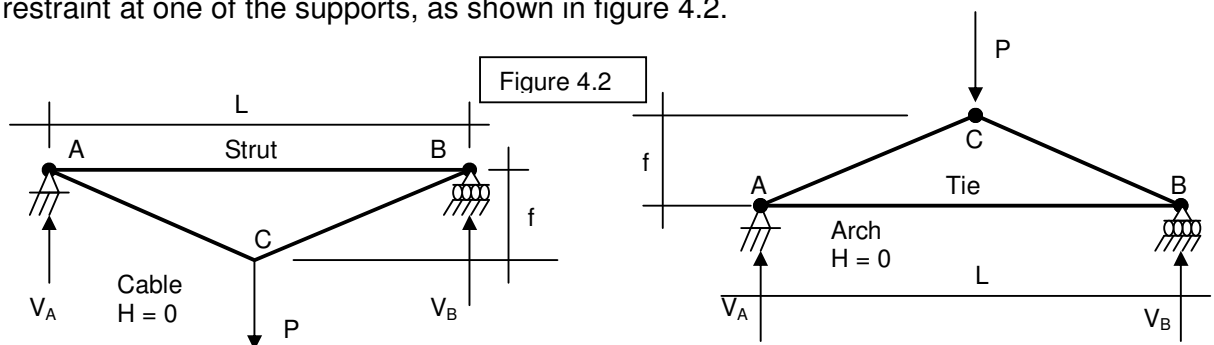
4.1.1 Origins of the Truss Structure

Consider the cable and the arch shown in figure 4.1 below.



The cable will carry the applied load by developing uni-axial **tension** in the portions CA and CB. The arch will carry the applied load by developing uni-axial **compression** in the members CA and CB.

However, in both structures, since $L \gg f$, large horizontal support reaction components, **H**, are developed at the supports. This horizontal reaction force (which is undesirable) can be eliminated by introducing a member between A and B, and thereby releasing the horizontal restraint at one of the supports, as shown in figure 4.2.



In the case of the cable, the member AB must be able to resist a **compressive force** equal to H, and hence it is called a **strut**. In the case of the arch, the member AB must be able to resist a **tensile force** equal to H, and hence it is called a **tie**.

Such combination of **tensile members** and **compression members** in **triangulated frameworks** are called trussed structures, or simply **trusses**. However, the members must not be “flexible” like in the cable, but must be “rigid” bars like in the arch.

4.1.2 Definition

A **plane truss (2-D)** is a system of members, all lying in the same plane, and joined together in such a manner as to form a **stable framework**. A space truss is a system of members arranged in 3-dimensions and joined together in such a manner as to form a stable framework. Each member of the truss will ideally carry a **uni-axial force only** (either tension or compression)

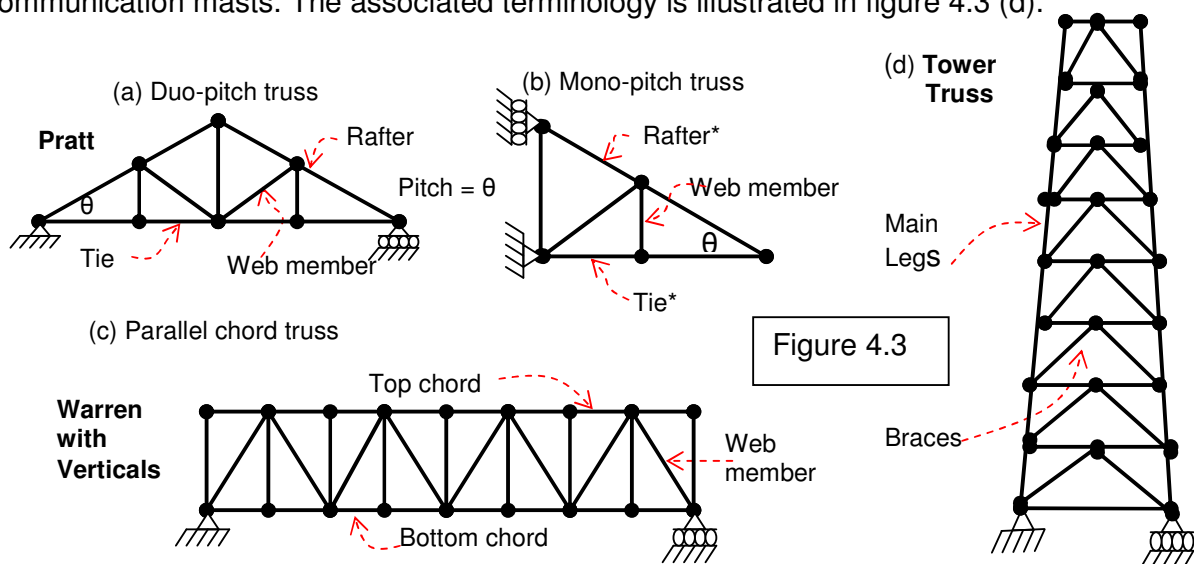
4.1.3 Applications

Trusses are commonly used in: -

Roofs, Bridges, Cranes, Chimney supports, Billboards, Elevated tanks, Communication masts, Power transmission masts, Scaffolding, etc

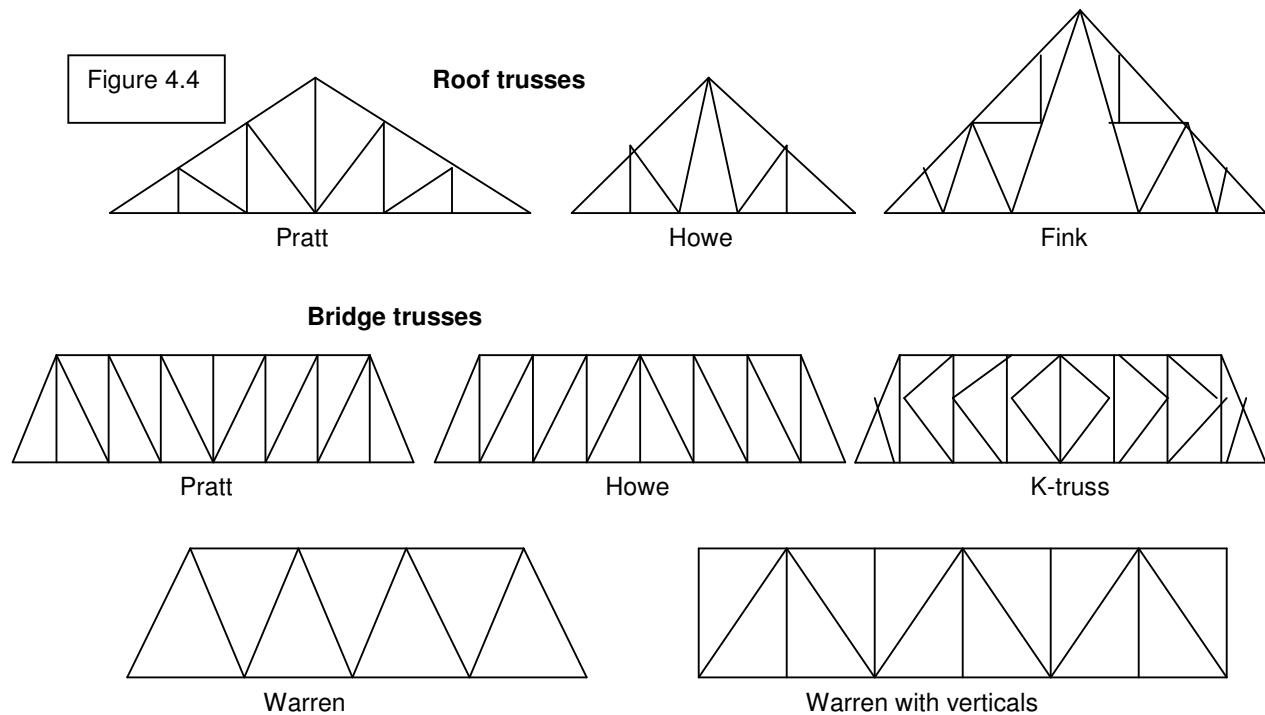
4.1.4 Associated Terminology

A **pitched truss** is one that is designed to drain water at a significantly large angle, called the **pitch**. There are both **duo-pitch** trusses and **mono-pitch** trusses as illustrated in figure 4.3 (a) and (b) below. The sloping members, which would normally carry compression forces, are called **rafters**. The horizontal member, which would normally carry tensile force, is called a **tie**. The members enclosed by the rafters and the tie will be either in tension or in compression, and are called **web members**. Note that when a mono-pitch truss acts as a cantilever, the rafter and tie interchange roles. A **parallel chord** truss is one which is designed to have minimum or no slope at all, as is common in bridge trusses. The associated terminology is illustrated in figure 4.3 (c). A **tower truss** is one which is designed to support loads at significantly high levels above the supports, as is common in communication masts. The associated terminology is illustrated in figure 4.3 (d).



4.1.5 Classical Types of Trusses

Some of the classical types of trusses are shown in figure 4.4 below.



Note that the engineer has full discretion as to the shape and/or geometry of any given truss. In practice, the number of shapes and geometry available is unlimited.

4.1.6 Static Determinacy

Considering a plane truss, let m = number of members, r = number of support reaction components, and j = number of joints.

Each member will carry a force which must be determined through analysis. The number of unknown forces is thus given by $m + r$. Each joint can be used to express two equations of equilibrium (i.e. $\sum F_x = 0$ and $\sum F_y = 0$). The number of available equations of equilibrium is thus equal to $2j$.

The degree of static indeterminacy is therefore given by: $\eta = m + r - 2j$

Note that for a space truss (3-D), the degree of static indeterminacy is given by:

$$\eta = m + r - 3j$$

4.1.7 Geometric Instability

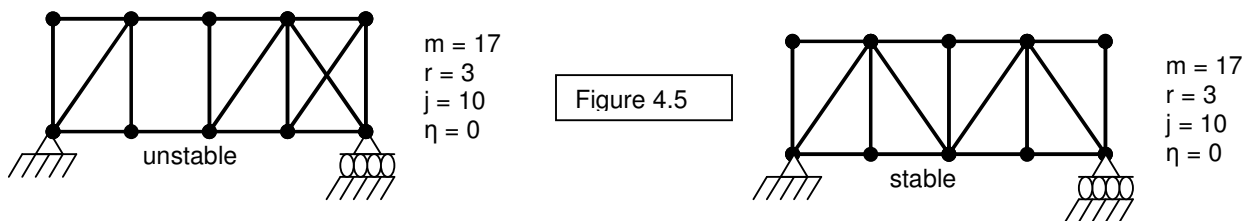
A geometrically unstable structure undergoes large displacements of rigid-body type before resisting the applied loading. A geometrically stable structure reacts spontaneously and instantaneously to resist the applied loading. Any displacements are small, and are due only to elastic straining. In the case of trusses, if: -

1. $\eta < 0 \Rightarrow$ the truss is unstable.
2. $\eta \geq 0 \Rightarrow$ the truss may or may not be stable.

Instability in trusses may be detected by **visual inspection**; by checking that: -

1. Support restraints are arranged in such a manner that the support reactions will neither be parallel nor concurrent.
2. The “space” enclosed by any group of members is triangular.

Figure 4.5 below illustrates instability in plane trusses where $\eta \geq 0$

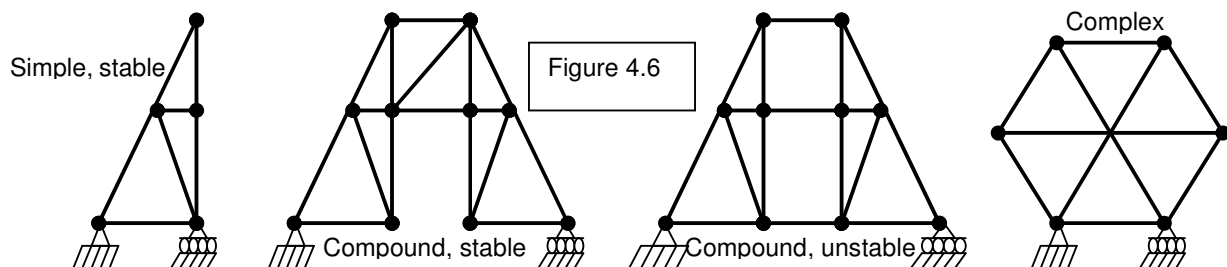


4.1.8 Classification of Plane Trusses

Plane trusses may be classified into three categories depending on their geometry: -

1. A **simple truss** is one that is made up of a series of triangular forms.
2. A **compound truss** is one that is made up by connecting two or more simple trusses in a stable manner.
3. A **complex truss** is one that cannot be classified as either simple or compound.

Figure 4.6 below illustrates these classes of plane trusses.



4.2 STRUCTURAL ANALYSIS

4.2.1 Objectives and Assumptions

Structural analysis of trusses is aimed at determining: -

5. The type, geometric stability and degree of static indeterminacy of the given truss.
6. The support reaction components,
7. The axial force induced in each member, and
8. The displacements of the joints of the truss

[Items 1, 2 and 3 are covered in FCE 201. Item 4 shall be covered in FCE 202]

The following **assumptions** are made in the idealization and analysis of plane trusses: -

1. All joints are **pin joints** (The joints are assumed to behave like frictionless hinges).
2. All member axes lie in the same plane, called the **middle plane**.
3. All external **loads** are point loads applied to the truss **at the joints** and acting in the middle plane.
4. The truss self weight is negligible compared to the applied loads.

4.2.2 Method of Equilibrium of Joints

The method is based on the fact that each joint must be in static equilibrium. This method is sometimes referred to simply as the **method of joints**. The procedure for this method may be outlined as follows: -

1. Evaluate the support reaction components.
2. For each joint: -
 - ✓ Isolate the joint as a fbd.
 - ✓ Write down the two equations of equilibrium.
 - ✓ Solve for the unknown member forces.
3. Next joint.
4. Present your Output using a fbd or using a table.

This procedure is illustrated in example 1 below.

Example 1

Determine the force induced in each member of the truss shown alongside.

Solution

Isolate the whole as a fbd

$$\sum F_x = H_A = 0$$

$$\sum F_y = V_A + V_B - 120 = 0$$

$$\sum M_A = 120 \times 4 - V_B \times 8 = 0$$

Solving: $V_A = 60 \text{ kN}$, $V_B = 60 \text{ kN}$, $H_A = 0$

Joint A

os $\theta = 4/5$ and $\sin \theta = 3/5$

$$\sum F_x = F_{AD} + F_{AC} \cos \theta = 0$$

$$\sum F_y = 60 + F_{AC} \sin \theta = 0$$

Solving: $F_{AC} = -100 \text{ kN}$, $F_{AD} = 80 \text{ kN}$

Joint D

$$\sum F_x = F_{BD} - 80 = 0$$

$$\sum F_y = F_{CD} - 120 = 0$$

Solving: $F_{BD} = 80 \text{ kN}$, $F_{CD} = 120 \text{ kN}$

Joint B

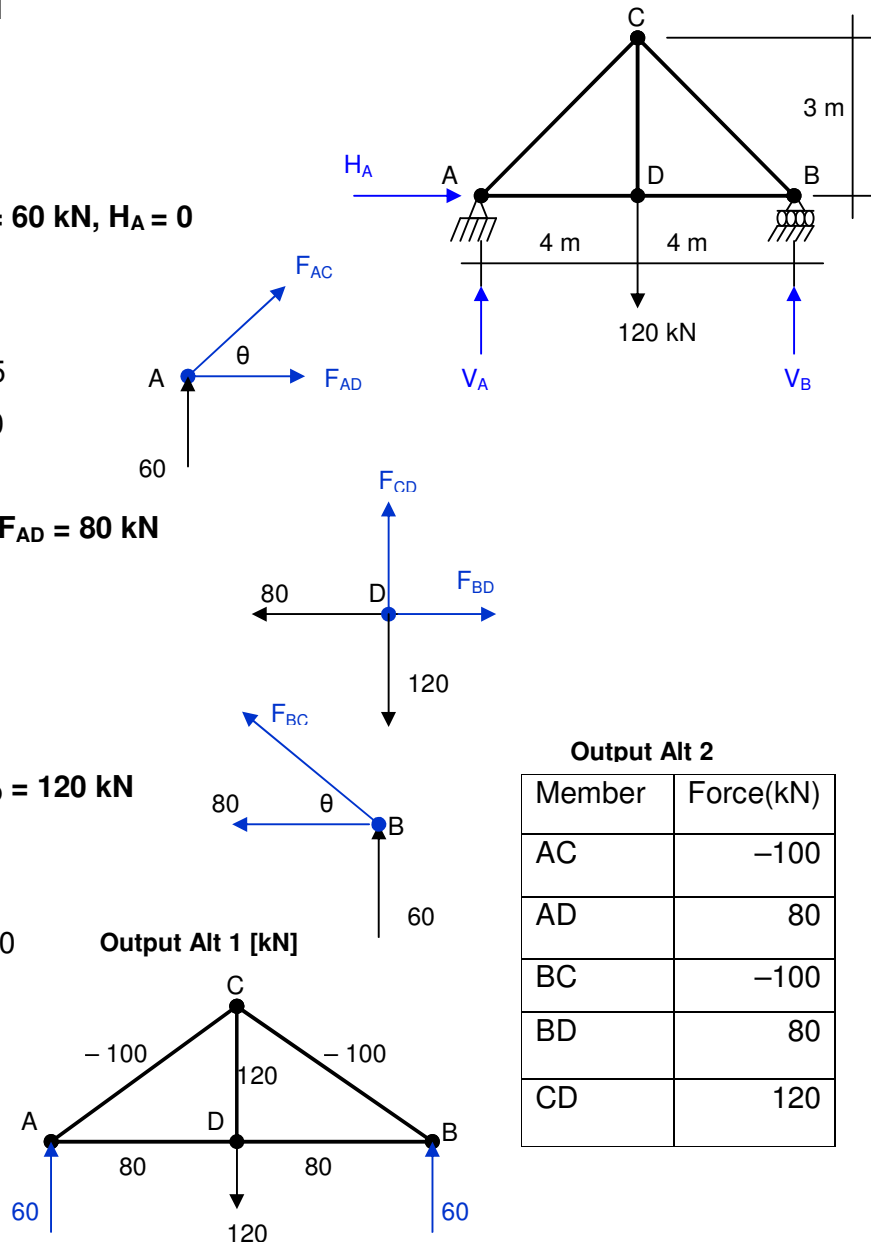
$$\sum F_x = -F_{BC} \cos \theta - 80 = 0$$

Solving: $F_{BC} = -100 \text{ kN}$

OUTPUT

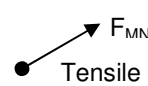
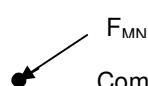
Alt 1 see **diagram**

Alt 2 see **table**



Notes

- The sign convention adopted is that a **tensile force is designated positive, while a compressive force is designated negative.**

- A tensile force is one that tends to pull the joint.  

A compressive force is one that tends to push the joint.

- Initially, any unknown member force is presumed to be tensile, and then solution of the equilibrium equations reveals its actual nature.
- The “starting joint” and the “next joint” in steps 2 and 3, respectively, of the above procedure should be such that there are no more than two unknown forces.
- Where possible take advantage of symmetry. [Full symmetry means symmetry with regard to the geometry of the structure, symmetry with regard to the arrangement of support restraints, and symmetry with regard to the applied loads].
- Suppose the 60 kN load in the above example was applied at C. [The solution is given in example 2

Example 2

Determine the force induced in each member of the truss shown alongside.

Solution

Isolate the whole as a fbd

$$\sum F_x = H_A = 0$$

$$\sum F_y = V_A + V_B - 120 = 0$$

$$\sum M_A = 120 \times 4 - V_B \times 8 = 0$$

Solving: $V_A = 60 \text{ kN}$, $V_B = 60 \text{ kN}$, $H_A = 0$

Joint A

$$\cos \theta = 4/5 \text{ and } \sin \theta = 3/5$$

$$\sum F_x = F_{AD} + F_{AC} \cos \theta = 0$$

$$\sum F_y = 60 + F_{AC} \sin \theta = 0$$

Solving: $F_{AC} = -100 \text{ kN}$, $F_{AD} = 80 \text{ kN}$

Joint D

$$\sum F_x = F_{BD} - 80 = 0$$

$$\sum F_y = F_{CD} = 0$$

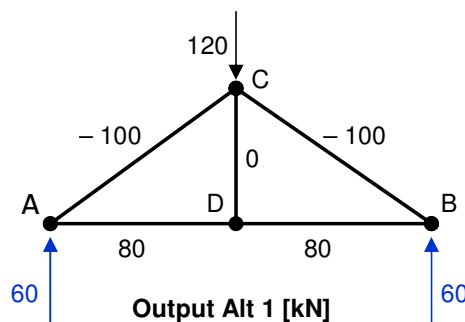
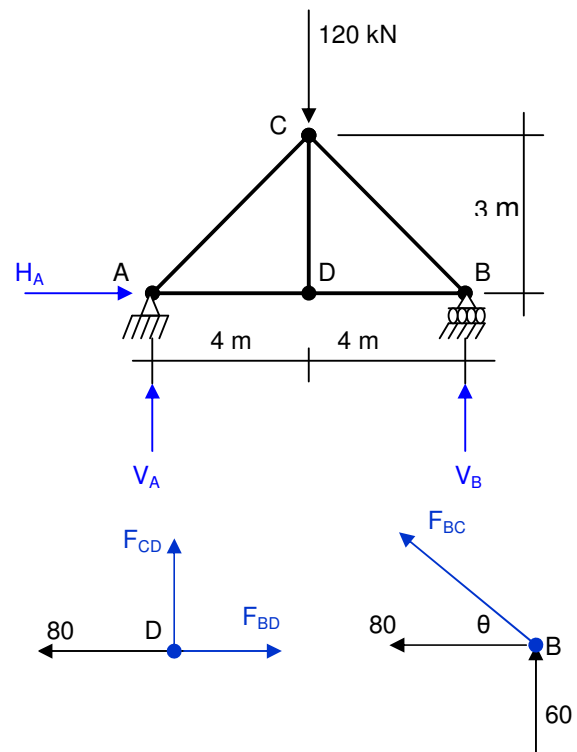
Solving: $F_{BD} = 80 \text{ kN}$,

$$F_{CD} = 0$$

Joint B

$$\sum F_x = -F_{BC} \cos \theta - 80 = 0$$

Solving: $F_{BC} = -100 \text{ kN}$



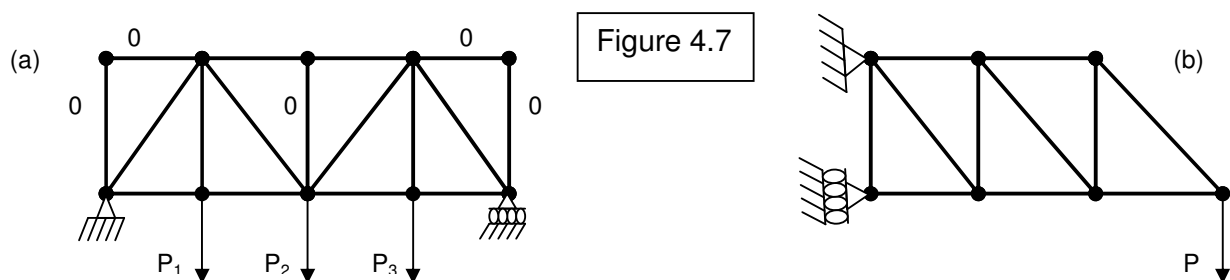
Output Alt 2

Member	Force(kN)
AC	-100
AD	80
BC	-100
BD	80
CD	0

Notes contd.

7. If there be a joint at which only three members meet, and if two of the members are collinear, then the force in the third member must be zero, provided that the joint is not externally loaded.
8. Similarly, if there be a joint at which only two members meet, which members are not collinear, then the force in each of the members must be zero, provided that the joint is not externally loaded. Figure 4.7 (a) illustrates the above two cases.

[Such zero load members should be identified before any analysis is done, and labeled zero]



9. In certain cases, evaluating support reaction components may not be a pre-requisite. One can go to step 2 of the procedure directly, and then the support reaction components can be determined like any other unknown force. This is because there exists a starting point, i.e. a joint where there are only two unknown forces. Figure 4.7 (b) illustrates one such case.

4.2.3 Method of Equilibrium of Sections

The method is based on the fact that any portion of the truss must be in static equilibrium. This method is sometimes referred to simply as the **method of sections**. Suppose only the forces in a few members of a truss are required. A **section** may be **imagined** through the members of interest. If the section reveals three non-parallel, non-concurrent forces, then these forces may be determined by applying the equation of static equilibrium.

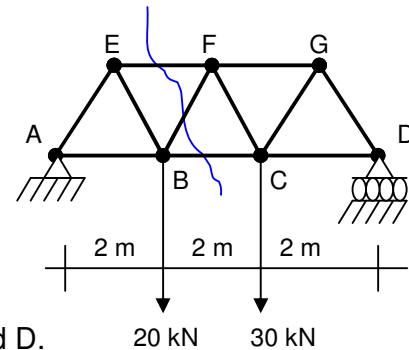
The procedure may be outlined as follows: -

1. Determine the support reaction components if necessary.
2. Imagine a section through the members of interest.
3. Sketch a fbd of the portion of the truss to one side of the imagined section.
4. Apply equilibrium to solve for the unknown forces.

Example 3 illustrates this procedure.

Example 3

Determine the member forces F_{EF} , F_{BF} , and F_{BC} of the truss shown below. All web members and rafters are inclined at 60° to the horizontal.



Solution

First determine the support reaction components at A and D.

$$[H_A = 0, V_A = 23.33 \text{ kN}, V_D = 26.67 \text{ kN}]$$

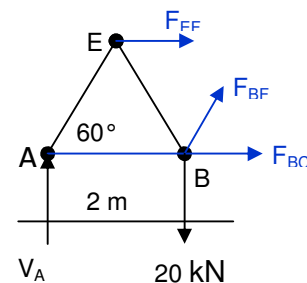
Next, imagine a section as shown in blue, and consider a fbd of the left hand portion of the truss.

$$\sum F_x = F_{EF} + F_{BC} + F_{BF} \cos 60^\circ = 0$$

$$\sum F_y = V_A - 20 + F_{BF} \sin 60^\circ = 0$$

$$\sum M_B = V_A \times 2 + F_{EF} (\tan 60^\circ) = 0$$

$$\text{Solving: } F_{BC} = 28.87 \text{ kN}, F_{BF} = -3.85 \text{ kN}, F_{EF} = -26.94 \text{ kN}$$



4.3 ANALYSIS OF COMPOUND TRUSSES

The analysis of compound trusses may require a combination of the method of joints and the method of sections. The general procedure may be outlined as follows: -

1. Identify the simple trusses forming the compound truss, and the manner in which they are connected (i.e. identify the connecting members and/or joints).
2. Imagine a section through the connecting members and/or joints, and isolate each simple truss as a fbd.
3. Apply equilibrium on the fbd(s) to determine the forces in the connecting members and/ or joints.
4. Use the method of joints to complete the analysis.

In some cases the method of joints alone will suffice. However, in the majority of cases the above procedure will apply. This is illustrated in figure 4.8 and 4.9 below.

Figure 4.8

Suppose it was required to determine the force induced in each member of the truss shown in figure 4.8.

It can be seen that although this is a compound truss the method of joints alone will suffice. This is because once the support reaction components are evaluated, one can start from, say joint A and go round all the other joints without encountering a joint where there are more than two unknown forces.

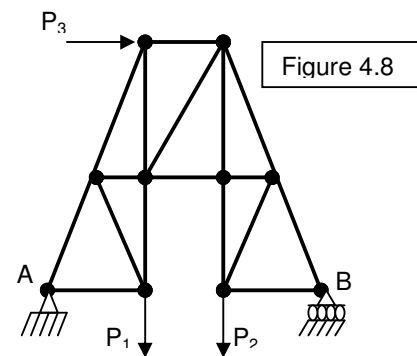


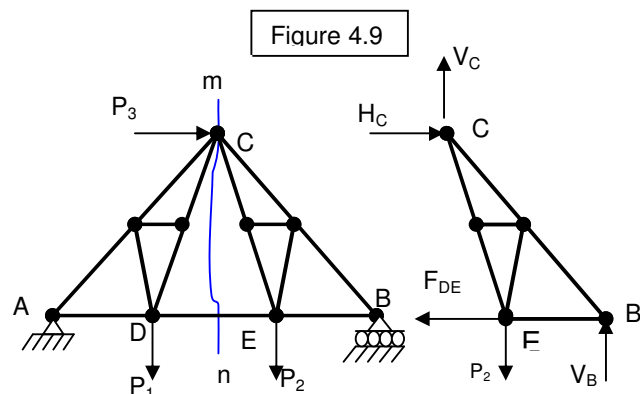
Figure 4.9

Suppose it was required to determine the force induced in each member of the truss shown in figure 4.9.

It can be seen that the method of joints alone cannot solve for all member forces in this truss.

The solution may then be achieved as follows: -

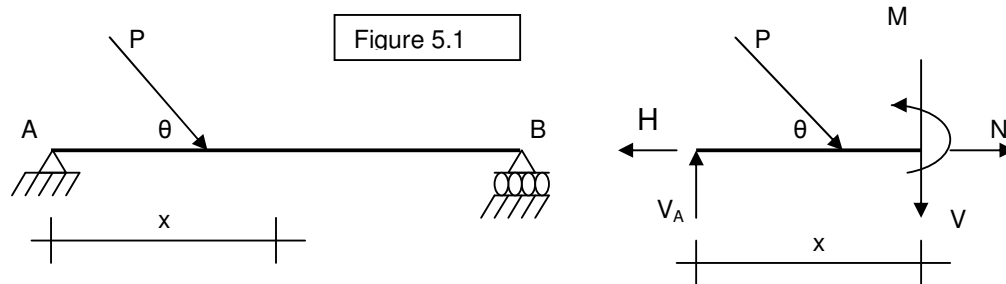
1. Determine support reaction components.
2. Imagine a section m-n as shown.
3. Using the fbd $\Sigma M_C = 0 \Rightarrow F_{DE}$ is obtained.
4. Use method of joints to complete the analysis.



5.0 BEAMS AND FRAMES

5.1 INTRODUCTION

Beams and **rigid jointed frames** are structural members which are predominantly subjected to **flexure** and are therefore referred to as **flexural structures**. Consider the simply supported beam shown in figure 5.1 below:-



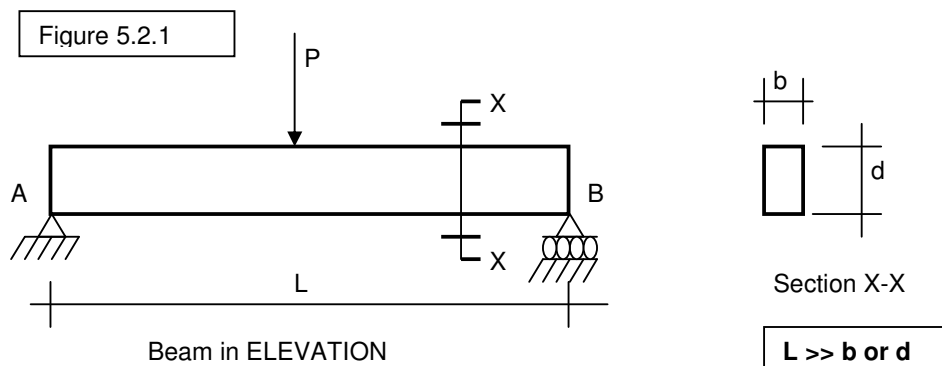
At any cross-section, say distance x from A, there will be **three resultant internal member forces**, namely, the **axial force N** , the **shear force V** , and the **bending moment M** . The effects of the axial force N are usually negligible compared to the effects of the shear force V and the bending moment M .

The techniques employed in the analysis of beams and frames are the same, and shall be introduced using the beam.

5.2 THE BEAM

5.2.1 Definition

A **shallow beam** (simply called a **beam**) is defined as a flexural member whose **span L is much greater than the cross-sectional dimensions** and which is subjected to transverse loads as illustrated in figure 5.2.1 below.

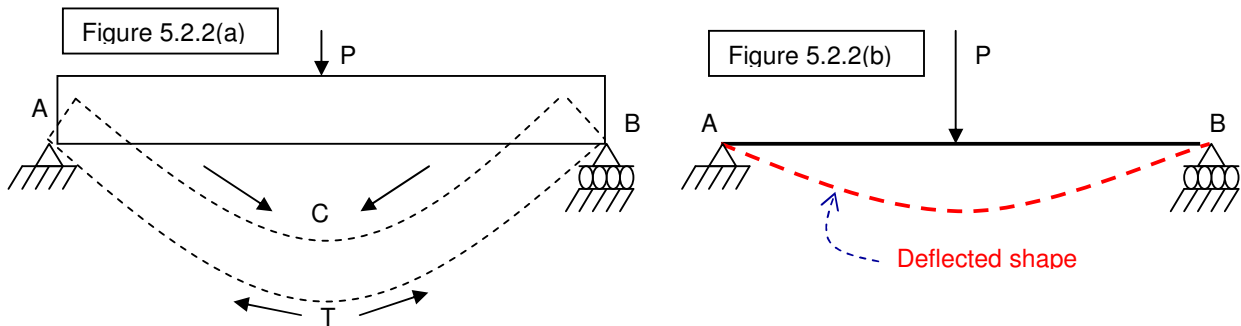


5.2.2 Mode of Action and Idealization

The beam carries the applied loads by changing its geometry from that of a **straight line** to that of a **curve** – a phenomenon referred to as **flexing**, the technical term for **bending**.

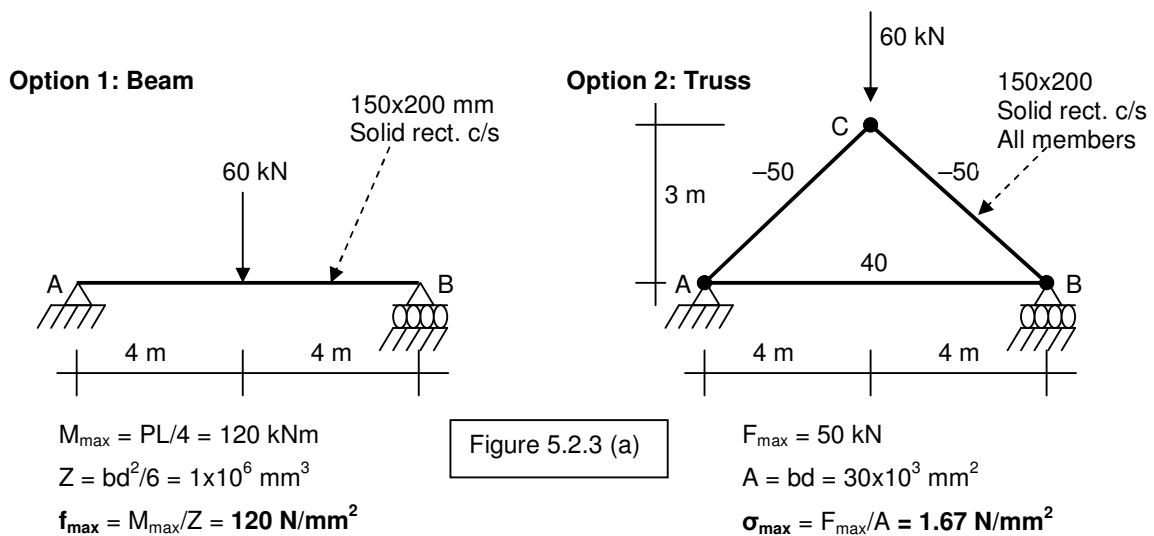
Flexure induces a **tension zone** in the bottom fibers and a **compression zone** in the top fibers, with a **neutral axis** dividing the two zones as illustrated in figure 5.2.2.(a).

The beam is idealized as a **line along its longitudinal axis** as illustrated in figure 5.2.2 (b) below.



5.2.3 The Beam Vs the Truss

Suppose you need to carry a 60 kN point load centrally placed over a span of 8 m. As a structural engineer you would have two options as illustrated in figure 5.2.3 below.



The maximum internal stresses induced in the beam and the truss (assuming there is no buckling) may be evaluated as shown in the figure.

The following deductions may be made by way of comparison:

Disadvantages of the beam

1. Very high stresses
2. Large flexural deflections

Advantages of the beam

1. Only one member was involved
2. No dead space
3. No buckling
4. Load can be placed anywhere

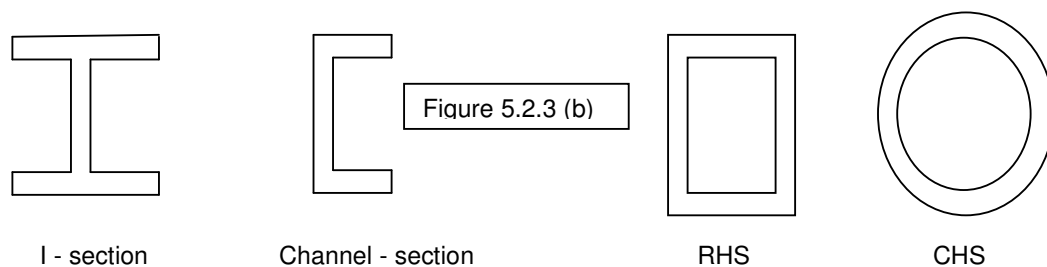
Advantages of the truss

1. Very low stresses
2. Small joint displacements

Disadvantages of the truss

1. Several members have to be used
2. Some dead space must be incurred
3. Buckling may occur
4. Load can only be placed at the joint C

The disadvantages of the beam listed above arise because the solid rectangular cross-section shape is not structurally efficient. In practice more structurally efficient sections are used, and the above disadvantages do not arise. The section shapes which are structurally efficient include those illustrated in figure 5.2.3 (b) below.



5.2.4 Static Determinacy and Stability

The degree of static indeterminacy of a beam depends on the arrangement of supports and the presence of internal hinges. Equilibrium of the whole beam can be expressed by the usual three equations. Each internal hinge facilitates the expression of an additional equation of equilibrium to define zero bending moment at the hinge. Thus, if r is the number of support reaction components, and i is the number of internal hinges, then the degree of static indeterminacy is given by $\eta = r - (3 + i)$

The stability of the beam is assessed based on the criteria set out in Chap 1 article 1.4.3, thus:- If $\eta > 0 \Rightarrow$ the beam is statically indeterminate; and the dsi $\eta = r - (3+i)$.

If $\eta = 0 \Rightarrow$ the beam is statically determinate; and the dsi $\eta = 0$.

If $\eta < 0 \Rightarrow$ the beam is **unstable!** [We do not deal with unstable structures]

5.2.5 Analysis of Beams

The structural analysis of beams involves the determination of:-

1. The **support reaction components**
2. The axial force **N**, shear force **V**, and bending moment **M**, at any given cross-section
3. The general nature of the deformed shape (termed the **deflected shape**)
4. The translational displacement (termed the **deflection δ**), and the rotational displacement (termed the **slope θ**) at any given cross-section.

Item no. 1 was covered in Chap 1 while item no. 4 will be covered in FCE 202. The purpose of this Chapter is to deal with item no. 2 and to develop ability to deal with item no. 3.

5.2.6 Axial Force, Shear Force and Bending Moment

The axial force **N**, shear force **V**, and bending moment **M** are **continuous (at least piece-wise) mathematical functions** over the span of the beam. These functions may be shown as **graphical variations** of the parameter under consideration, from cross-section to cross-section along the beam.

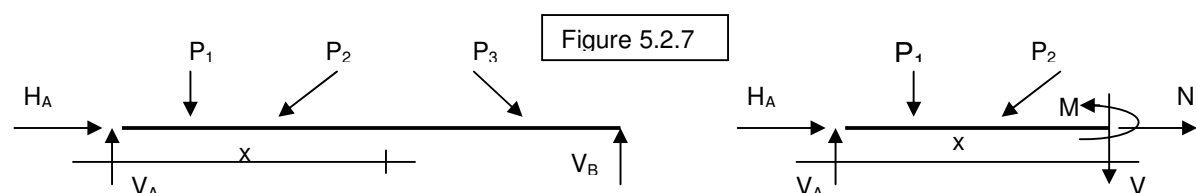
Such graphical representations are referred to as **diagrams**. We shall draw the axial force diagram (**AFD**), the shear force diagram (**SFD**), and the bending moment diagram (**BMD**).

However, the AFD may be omitted from the analysis for two reasons, namely: - The effects of axial stresses are negligible compared to the effects of shear and flexural stresses and the AFD may be null throughout the beam

5.2.7 Definitions

Consider the simply supported beam shown in figure 5.2.7 below. At any cross-section distance x from the left hand end A, we shall define:

1. Axial force **N** as the algebraic sum of all horizontal forces to the left of the cross-section.
2. Shear force **V** as the algebraic sum of all vertical forces to the left of the cross-section.
3. Bending moment as the algebraic sum of all bending moments due to all vertical forces to the left of the cross-section.

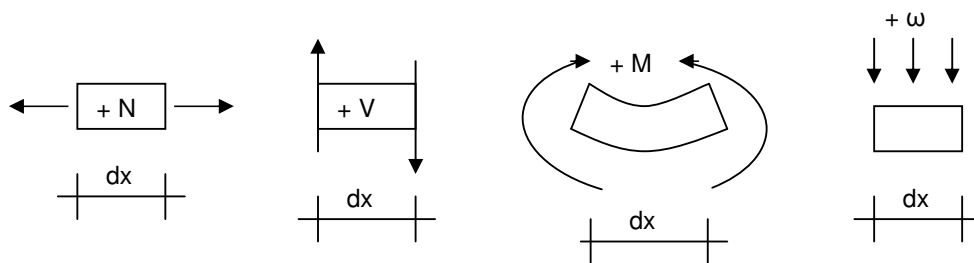


5.2.8 Sign Convention

Considering an elemental portion of the beam, of length dx , and situated immediately to the left of the cross-section X , the following sign convention shall be adopted as illustrated in figure 5.2.8 below.

1. **Positive axial force** is that which tends to cause **tensile stress**.
2. **Positive shear force** is that which tends to cause **clockwise relative rotation**.
3. **Positive bending moment** is that which tends to cause **sagging**.
4. **Deflection** is measured positive downwards.
5. **Slope** is measured positive clockwise.
6. **Positive applied loads** are those that act **downwards**

Figure 5.2.8

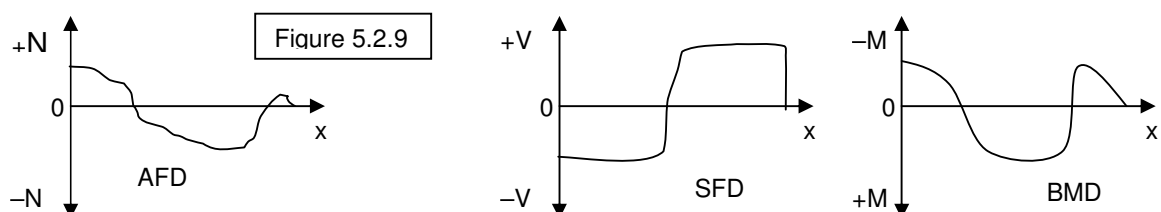


5.2.9 Plotting Convention

The longitudinal axis of the beam is taken as the base line which coincides with the x -axis whose origin is taken as the left hand end of the beam. The various load effects are then plotted using the following convention:-

1. **Positive axial force** is plotted **above** the base line.
2. **Positive shear force** is plotted **above** the base line.
3. **Positive bending moment** is plotted **below** the base line.

Figure 5.2.9.below illustrates the convention using arbitrarily scribbled diagrams.



For the words “above” and “below” to have meaning, the beam is assumed to be a **normal beam** (i.e. a horizontal beam). If the beam is not horizontal then it is treated as a frame, see section 5.5.

5.2.10 Plotting SFD, BMD, D-Shape: First Principles

The solution procedure **using first principles** may be outlined as follows (see example 1):-

1. Determine the support reaction components and draw a fbd of the whole beam
2. Identify the **regions** making up the beam. (The regions are defined by the existing boundary conditions i.e points of application of the point loads, or support points, or internal hinge, etc. In each region, the various load effects will be continuous).
3. For each region:-
 - ✓ Consider a fbd of a portion of the beam from the LH end to somewhere in that region as illustrated in figure 5.2.7
 - ✓ Use equilibrium to find mathematical expressions for N, V, and M
 - ✓ Use these expressions to plot the load effects using the conventions developed above (use a suitable “scale” to give clear picture of the diagrams)
4. Next region
5. Close each diagram (i.e. make each diagram to start and end at zero).
6. Sketch the deflected shape.
7. STOP: END.

Example 1

- (a) Draw the AFD, SFD and BMD for the simply supported beam shown below.
- (b) Sketch the deflected shape.

Solution

Determine support reaction components and resolve inclined forces to obtain the fbd shown. The beam has regions AB, BC, and CD.

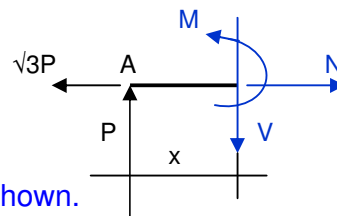
Region AB: $0 \leq x < L/3$

$$\sum F_x = -\sqrt{3}P + N = 0 \Rightarrow N = \sqrt{3}P \text{ (constant)}$$

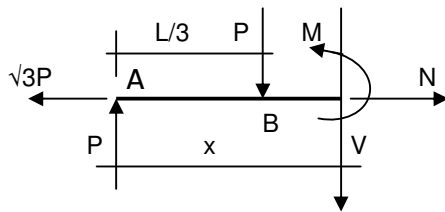
$$\sum F_y = P - V = 0 \Rightarrow V = P \text{ (constant)}$$

$$\sum M = Px - M = 0 \Rightarrow M = Px \text{ (linear)}$$

These “graphs” are then plotted in the region AB as shown.



Region BC: $L/3 \leq x < 2L/3$



$$\sum F_x = -\sqrt{3}P + N = 0 \Rightarrow N = \sqrt{3}P \text{ (constant)}$$

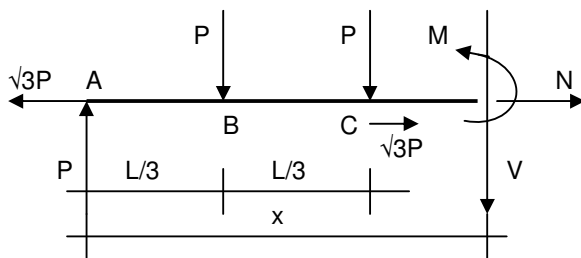
$$\sum F_y = P - P - V = 0 \Rightarrow V = 0$$

$$\sum M = Px - P(x - L/3) - M = 0$$

$$\Rightarrow M = PL/3 \text{ (constant)}$$

These “graphs” are then plotted in the region BC as shown.

Region CD: $2L/3 \leq x < L$



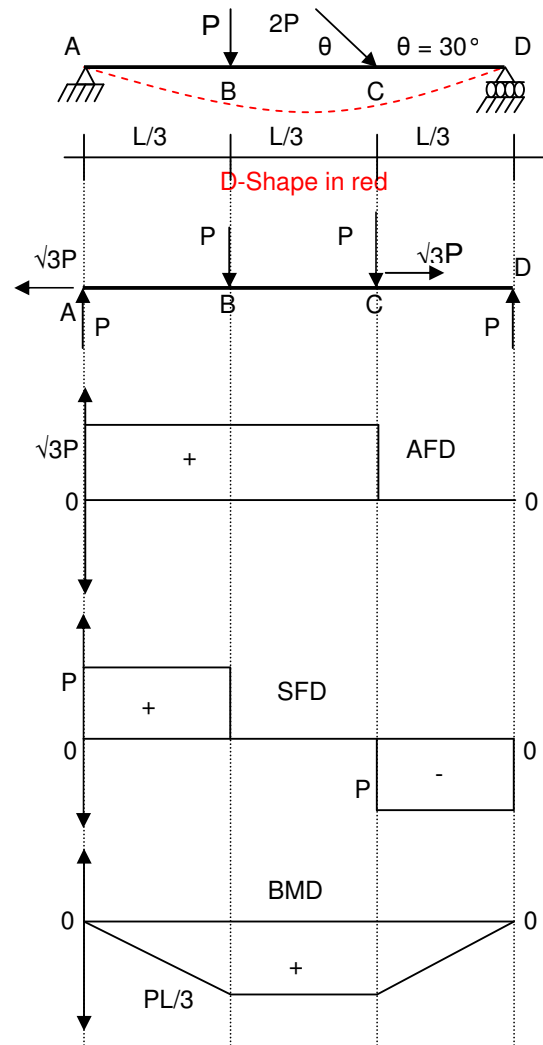
$$\sum F_x = -\sqrt{3}P + \sqrt{3}P + N = 0 \Rightarrow N = 0$$

$$\sum F_y = P - P - P - V = 0 \Rightarrow V = -P$$

$$\sum M = Px - P(x - L/3) - P(x - 2L/3) - M = 0$$

$$\Rightarrow M = P(L - x) \text{ (linear)}$$

These “graphs” are then plotted in the region CD as shown.



Notes:

1. The SFD “seems” to follow the pattern of point loads acting on the fbd.
2. The mathematical expressions for V and M, in each region, “seem” to be related.
3. The mathematical expressions for N have no relation with those for V and M.
4. The deflected shape is plotted as shown in red.
5. Although the BMD may be kinked, the D-shape is a continuous curve.
6. The D-shape corresponds to the BMD in the sense that positive moment is plotted on the tension side.

Example 2

- (a) Draw the SFD and BMD for the simply supported beam shown below
- (b) Sketch the deflected shape

Solution

Determine support reaction components to obtain the fbd shown.

The beam has only one region AB.

Region AB: $0 < x < L$

$$\sum F_y = \omega L/2 - \omega x - V = 0$$

$$\Rightarrow V = \omega(x - L/2) \text{ a linear function.}$$

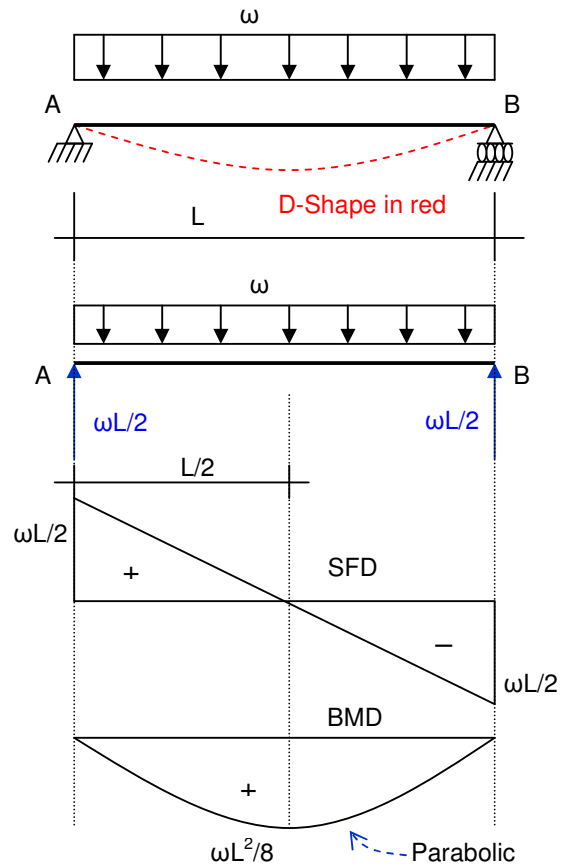
$$\sum M = \omega Lx/2 - \omega x^2/2 - M = 0$$

$$\Rightarrow M = (\omega/2)(Lx - x^2) \text{ a parabolic function.}$$

When $x = 0$ or $x = L$, $M = 0$

and when $x = L/2$, $M = \omega L^2/8$ the max value.

These “graphs” are then plotted in the region AB as shown.



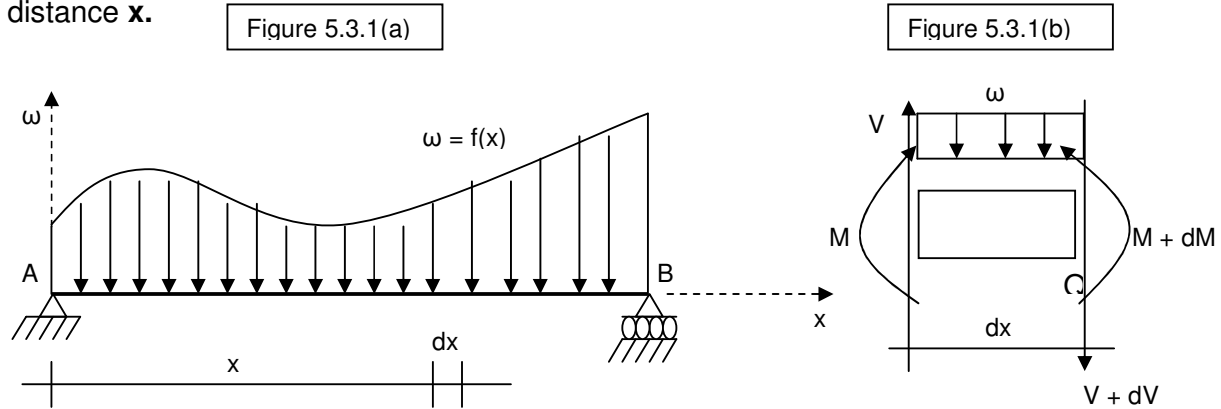
Notes:

1. Normal beams are usually subjected to transverse loads only. The axial force N is zero throughout the beam, and the AFD may be omitted.
2. An inspection of the mathematical expressions for V and M reveals that these load effects are related. This relation forms the subject for section 5.3 below.
3. After a little experience, it may not be necessary to show the zeros on the baseline or the arrows on the ordinate line.

5.3 LOAD – SHEAR – MOMENT RELATIONS

5.3.1 Distributed Loads

Consider the simply supported beam shown in figure 5.3.1(a) below, which is subjected to a general distributed load ω , whose intensity at any point is a mathematical function of the distance x .



Consider an element of the beam at a distance x measured from the left hand end A, and of elemental length dx where dx is approaching zero. Such an element is shown in figure 5.3.1(b) using an exaggerated scale only for purposes of clarity.

It shall be assumed that:

1. Within the distance dx , the load $\omega(x)$ remains constant.
2. Second order differential quantities are negligible.

Equilibrium of the above element requires that:-

$$\sum F_y = V - \omega dx - (V + dV) = 0$$

$$\Rightarrow \quad dV/dx = -\omega \quad \text{-----} (1)$$

The rate of change of shear force w.r.t. x is equal to the applied load in magnitude, but opposite in sense.

$$\sum M_Q = M + Vdx - \omega dx dx/2 - (M + dM) = 0$$

$$\Rightarrow \quad dM/dx = V \quad \text{-----} (2)$$

The rate of change of bending moment w.r.t. x is equal to the shear force both in magnitude and sense.

5.3.2 Interpretation and Application of Eqns 1 and 2

Consider equation (1): $dV/dx = -\omega$; this equation means that in any region: -

- ✓ If the load ω is zero, then the shear force V will be constant.
- ✓ If the load ω is constant, then the shear force V will be linear.
- ✓ If the load ω is linear, then the shear force V will be parabolic, and so on.

Thus the shear force will always be one order higher than the applied load.

Equation (1) may also be expressed as $dV = -\omega dx$, which then means that:

$$\Delta V = - \int \omega dx \text{ ----- (1*)}$$

The change in shear force from, say, point A to point B along the beam, is equal in magnitude but opposite in sense to the area under the loading diagram between the points A and B.

Consider equation (2): $dM/dx = V$; this equation means that in any region:-

- ✓ If the shear force V is zero, then the bending moment M will be constant.
- ✓ If the shear force V is constant, then the bending moment M will be linear.
- ✓ If the shear force V is linear, then the bending moment M will be parabolic, etc.

Thus the bending moment will always be one order higher than the shear force.

Equation (2) may also be expressed as $dM = Vdx$, which then means that:

$$\Delta M = \int Vdx \text{ ----- (2*)}$$

The change in bending moment from, say, point A to point b along the beam, is equal (both in magnitude and sense) to area under the shear force diagram between the points A and B.

Notes:

1. Maximum and/or minimum bending moment will occur when $dM/dx = 0$, that is when the shear force V crosses the baseline. However maximum and/or minimum bending moment may also occur where there is an applied concentrated moment.
2. Equations (1) and (2) may be combined thus:

$$dV/dx = -\omega \quad \text{and} \quad dM/dx = V \quad \text{so that} \quad d/dx[dM/dx] = -\omega \quad \text{which gives:}$$

$$d^2M/dx^2 = -\omega \text{ ----- (3)}$$

Equation (3) is the basis of beam deflection theory which is covered in FCE 202.

5.3.3 Point Load

Consider the point load P acting at the point C of the simply supported beam shown in figure 5.3.3(a). A fbd of the cross-section at C is isolated and shown in figure 5.3.3(b), where the dimension dx is ideally equal to **zero**, but has been blown out only for purposes of clarity. In figure 5.3.3(b), V_L and M_L are the shear force and bending moment, respectively, just to the left of the point C . V_R and M_R are defined similarly just to the right of the point C .

Equilibrium of the c/s section at C requires that:

$$\sum F_y = V_L - P - V_R = 0$$

$$\Rightarrow V_R = V_L - P$$

The shear force just to the right is equal to the shear force just to the left less the point load.

Conclusion: A point load causes a first order discontinuity (called a jump) on the SFD.

Similarly:

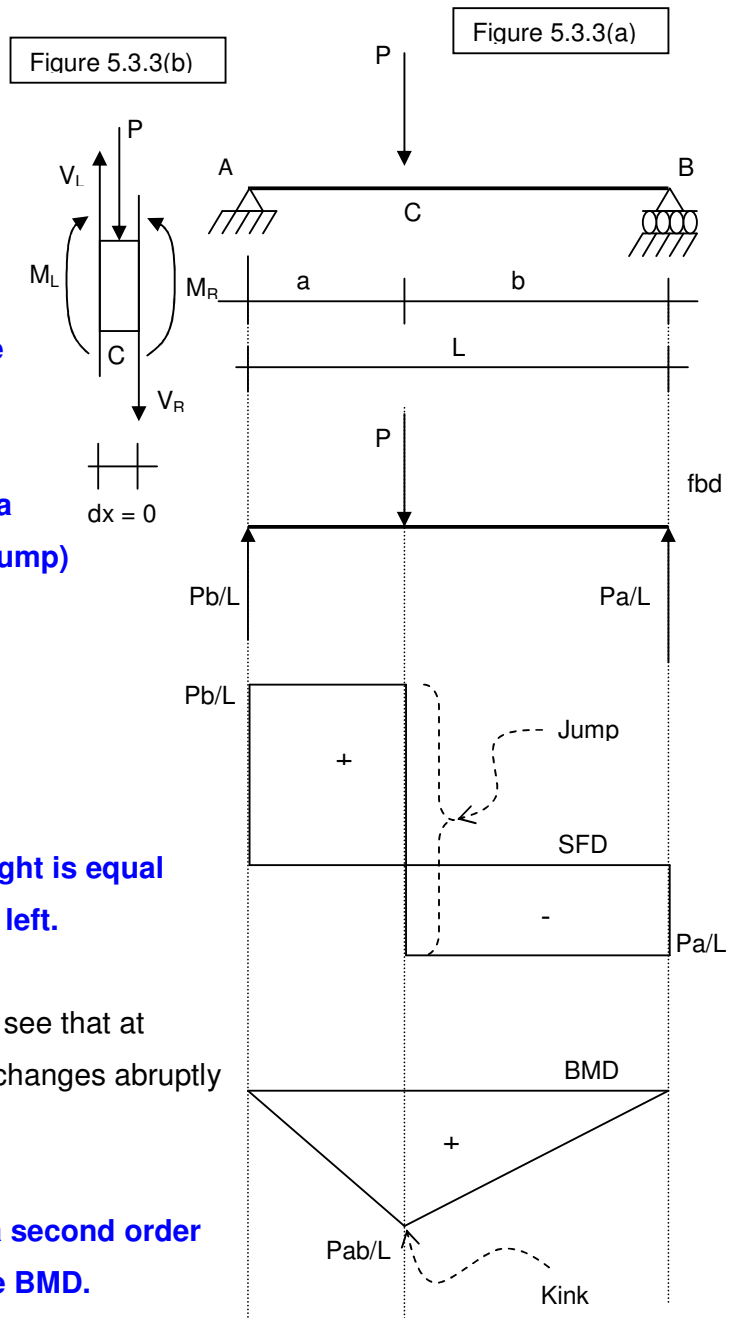
$$\sum M_C = M_L - M_R = 0$$

$$\Rightarrow M_L = M_R$$

The bending moment just to the right is equal to the bending moment just to the left.

However, by inspecting the BMD we see that at the point C the gradient of the BMD changes abruptly from positive to negative.

Conclusion: A point load causes a second order discontinuity (called a kink) on the BMD.



5.3.4 Concentrated Moment

Consider the concentrated moment M acting at the point C of the simply supported beam shown in figure 5.3.4(a). A fbd of the cross-section at C is isolated and shown in figure 5.3.4(b), where the dimension dx is ideally equal to zero, but has been blown out only for purposes of clarity. In figure 5.3.4(b), V_L and M_L are the shear force and bending moment, respectively, just to the left of the point C . V_R and M_R are defined similarly just to the right of the point C .

Equilibrium of the c/section at C requires that:

$$\sum F_y = V_L - V_R = 0 \Rightarrow V_R = V_L$$

The shear force just to the right is equal to the shear force just to the left.

Conclusion: A concentrated moment has no effect on the SFD.

Similarly:-

$$\sum M_C = M_L - M_R + M = 0 \Rightarrow M_R = M_L + M$$

The bending moment just to the right is equal to the bending moment just to the left plus the applied concentrated moment.

Conclusion: A concentrated moment causes a first order discontinuity (called a jump) on the BMD.

Figure 5.3.4(b)

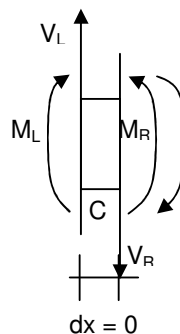
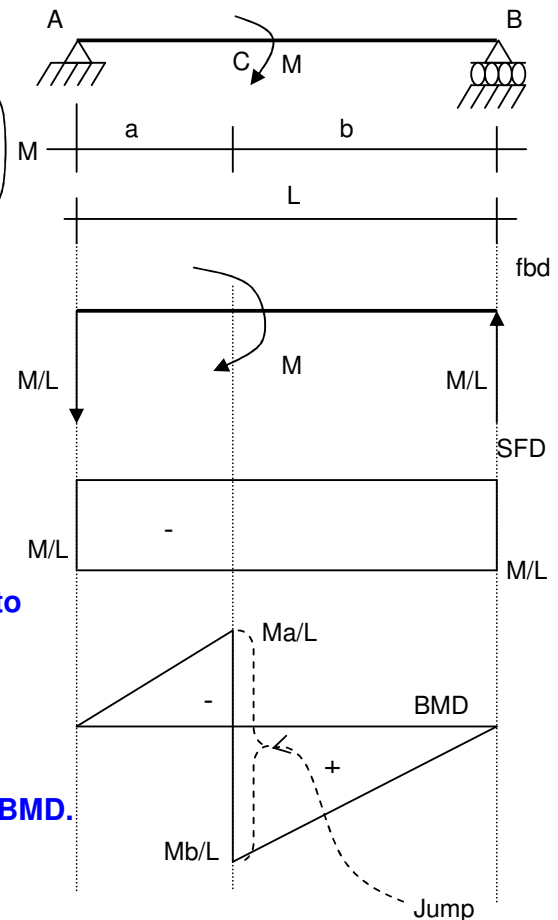


Figure 5.3.4(a)

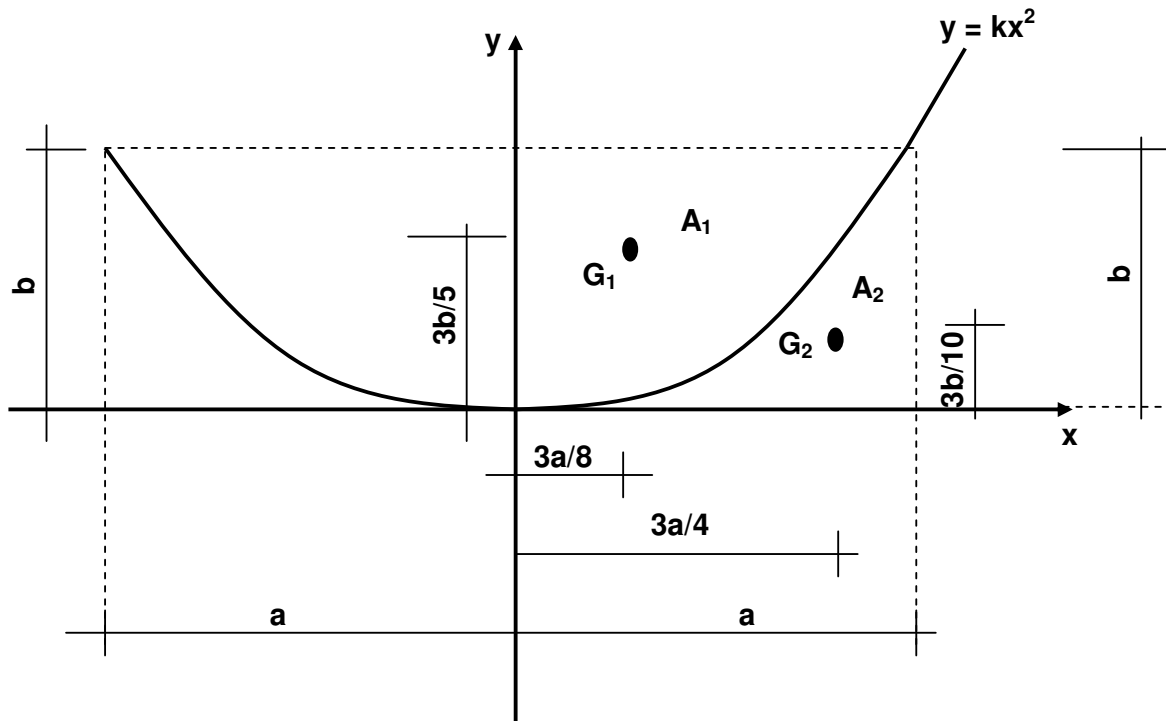


NOTE

The interpretations, observations and conclusions outlined in articles 5.3.2 to 5.3.4 are used to plot the SFD and BMD using second principles. The SFD is generated from the loading diagram (which is given), then the BMD is generated from the SFD. All one needs to know is what is happening at point “A”, and then predict what is happening at point “B”. In doing so, the area under the relevant diagram has to be calculated. This area can be calculated using first principles (integration) or second principles (Handout # 1).

HANDOUT # 1

AREAS UNDER A PERFECT PARABOLIC FUNCTION



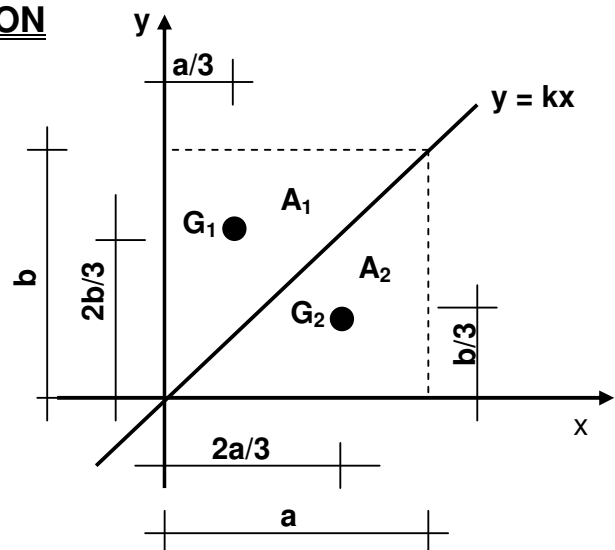
$$A_1 = (2/3)ab$$

$$A_2 = (1/3)ab$$

$$G_1 (3a/8, 3b/5)$$

$$G_2 (3a/4, 3b/10)$$

AREAS UNDER A LINEAR FUNCTION



$$A_1 = A_2 = \frac{1}{2} ab$$

$$G_1 (\frac{1}{3} a, \frac{2}{3} b)$$

$$G_2 (\frac{2}{3} a, \frac{1}{3} b)$$

5.3.5 Plotting SFD, BMD, D-Shape: Second Principles

The solution procedure **using second principles** may be outlined as follows:-

1. Determine the support reaction components and sketch the **load diagram** (a fbd of the whole beam).
2. Generate the **SFD** from the load diagram.
3. Generate the **BMD** from the SFD.
4. Generate the **D-shape** from the BMD.

5.4 SKETCHING DEFLECTED SHAPE

The deflected shape is usually sketched against the given structure (either in a different color or in pencil)

The deflected shape closely corresponds to the BMD. By our convention, positive bending moments are plotted below the baseline. This ensures that the BMD corresponds to the tension side of the beam:

1. When the BMD is positive, tension is on the bottom fibers of the beam, and the beam is sagging.
2. When the BMD is negative, tension is on the top fibers of the beam, and the beam is hogging.
3. When the BMD incurs a point of contra-flexure, the deflected shape incurs a point of inflection.
4. Although the BMD may have kinks or jumps, the deflected shape must be a mathematically continuous smooth curve.

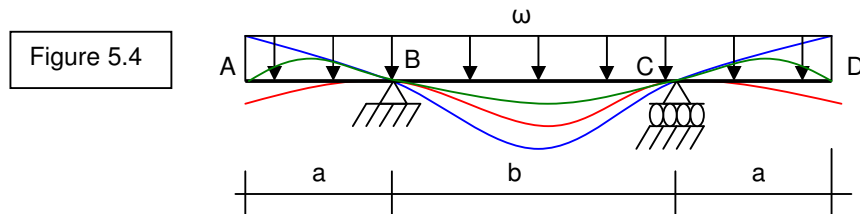
The procedure is based on the above correspondence, and a bit of common sense.

However, one has to ensure that the deflected shape satisfies all existing boundary conditions. For example:-

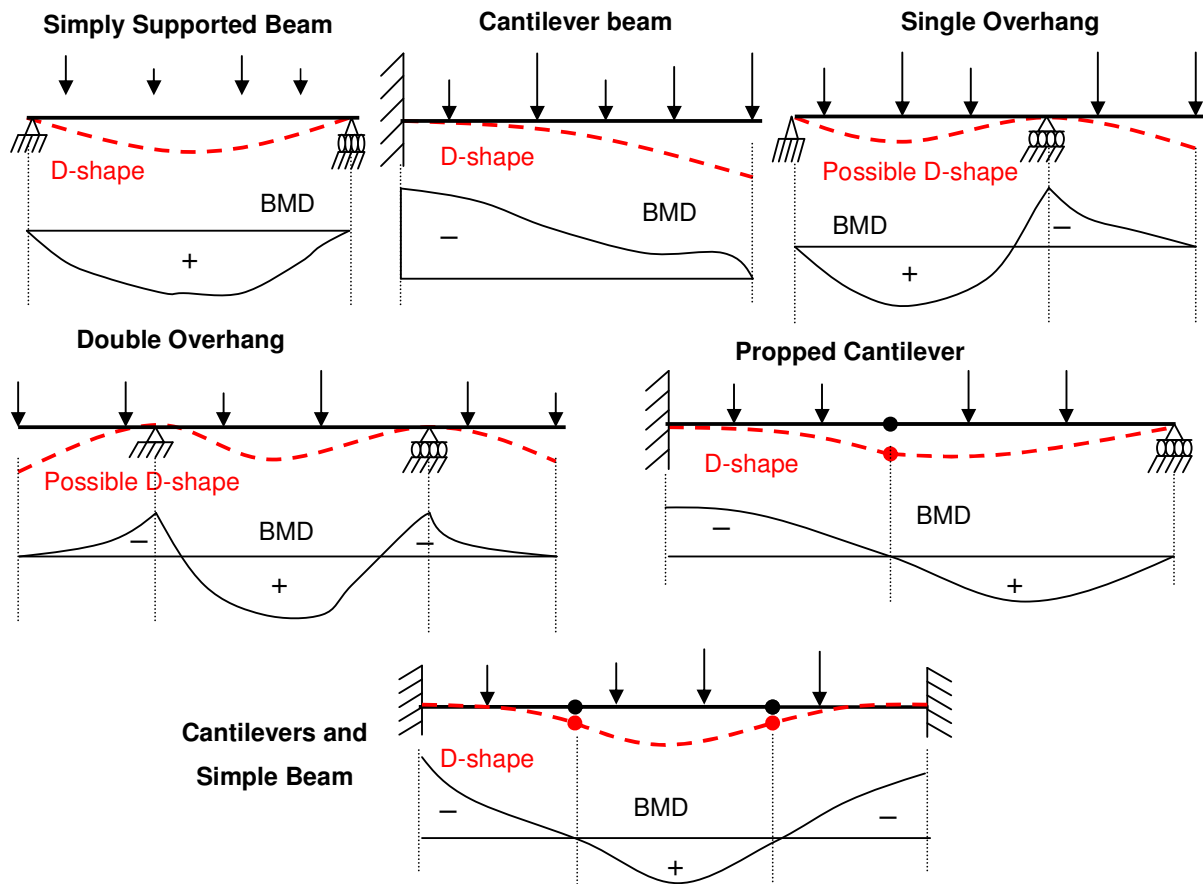
1. The deflected shape must approach a fixed support at zero slope w.r.t. the original beam orientation.
2. The deflected shape can have any slope over a pin or roller support, but cannot fly away from the support.
3. However an internal hinge allows free rotation, and therefore violates the above continuity requirement. A kink may therefore occur at an internal hinge.

The general behavior of a beam between any two boundary conditions is similar to the behavior of a simple beam with the same loading and span. The effect of internal supports is to incline the baseline of the equivalent simple beam. The behavior of some simple beams is illustrated in **Handout # 2**.

The deflected shape, at the level of FCE 201, remains a sketch and is not definite. For example any of the three deflected shapes for the beam shown in figure 5.4 below would be possible. The definite deflected shape can only be known when one is able to determine the actual deflections of the points A and D, which ability is given by FCE 202.



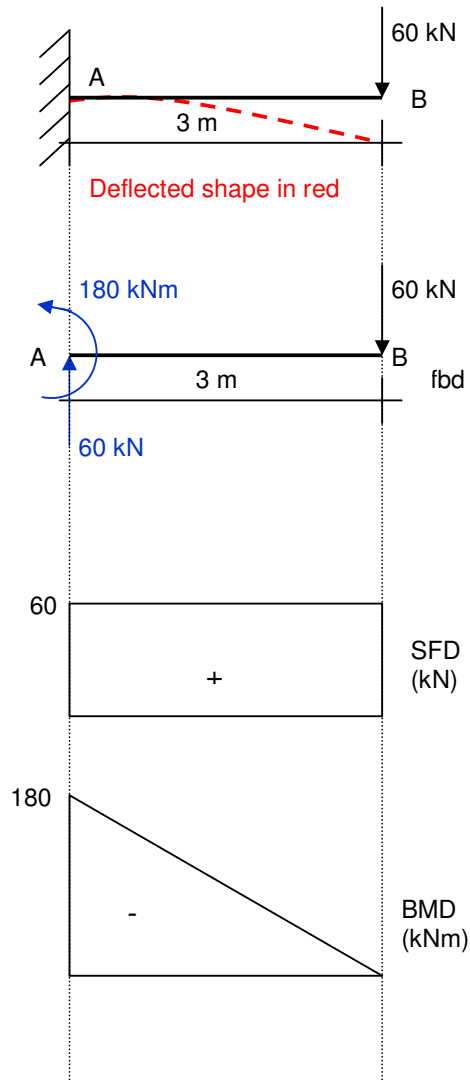
HANDOUT # 2: SIMPLE BEAM BEHAVIOR



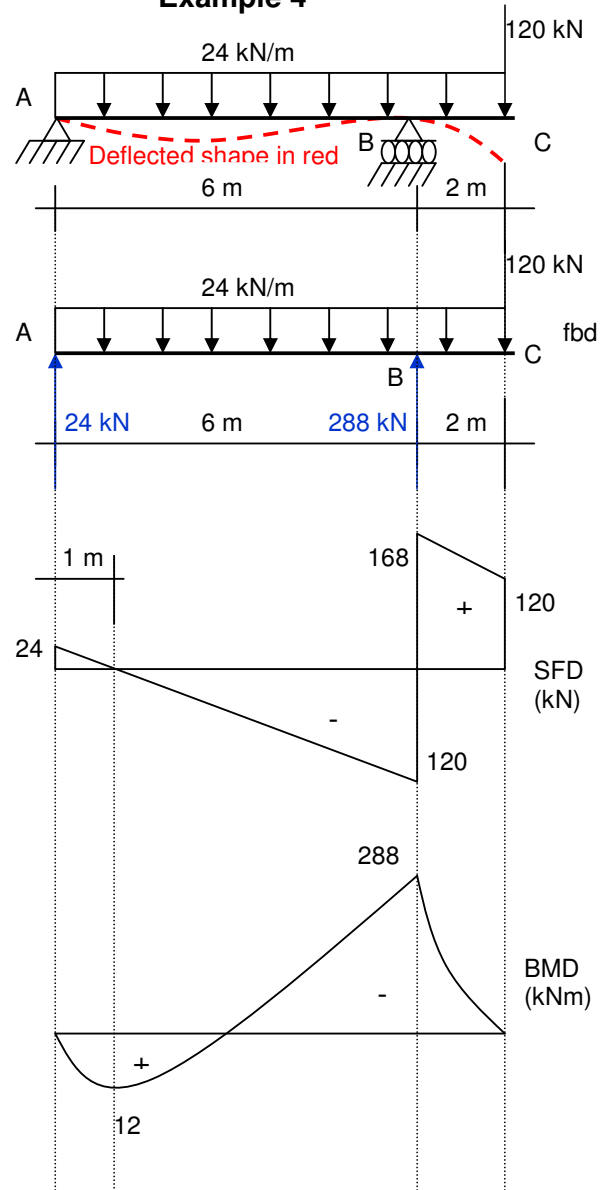
Worked Examples

Draw the SFD and BMD and sketch the D-shape for each of the beams shown in examples 3 to 8

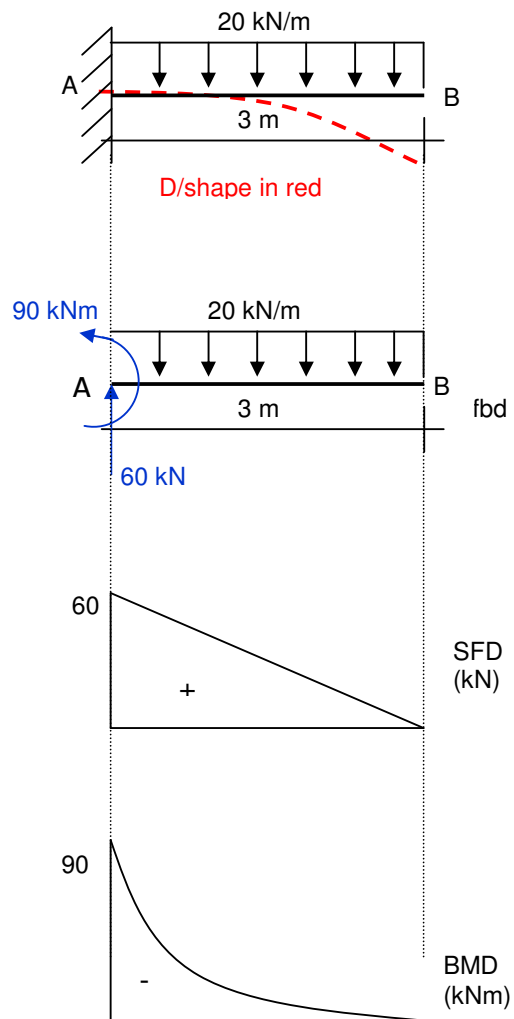
Example 3



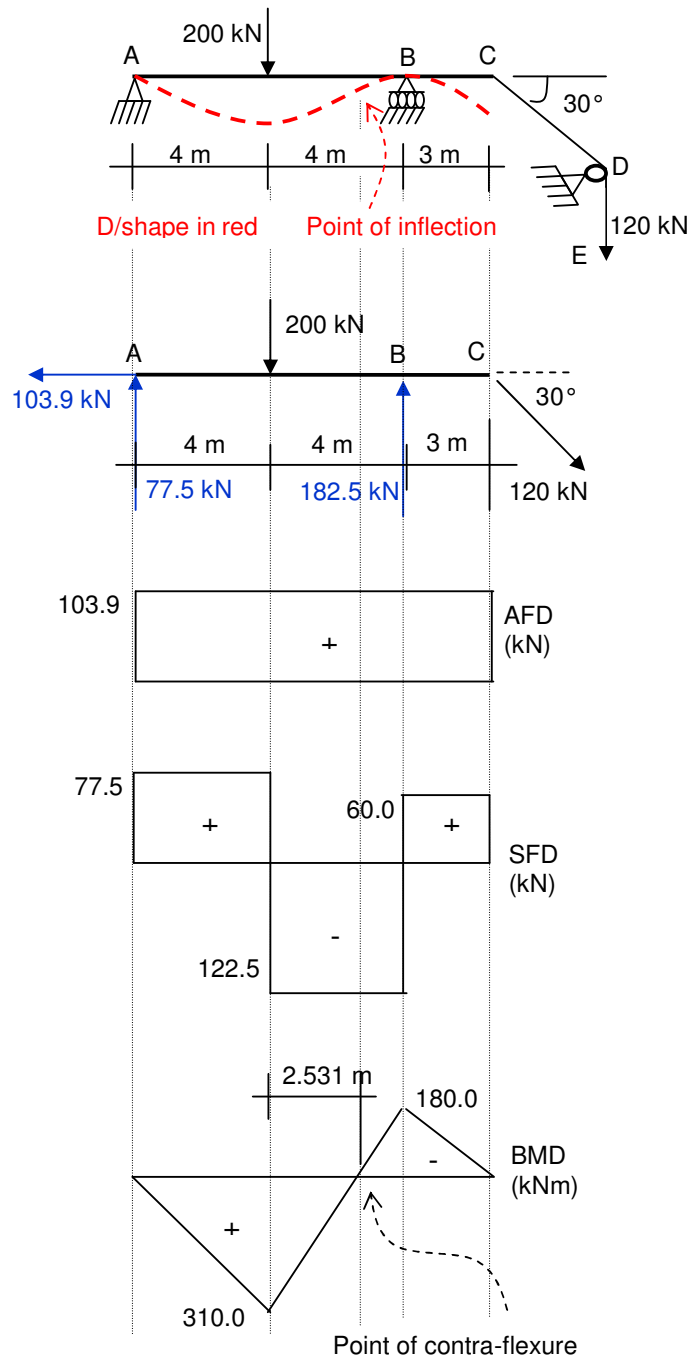
Example 4



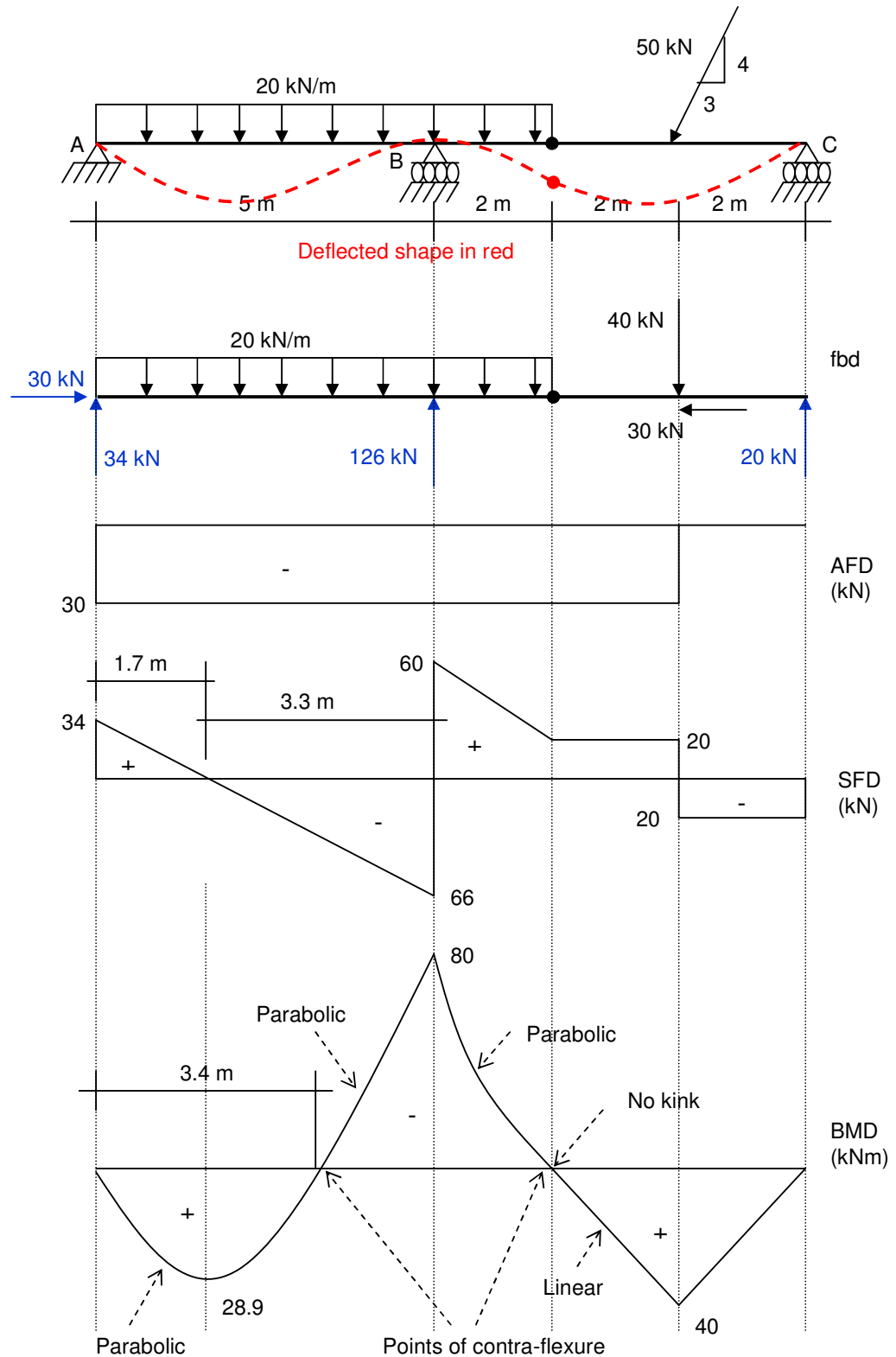
Example 5



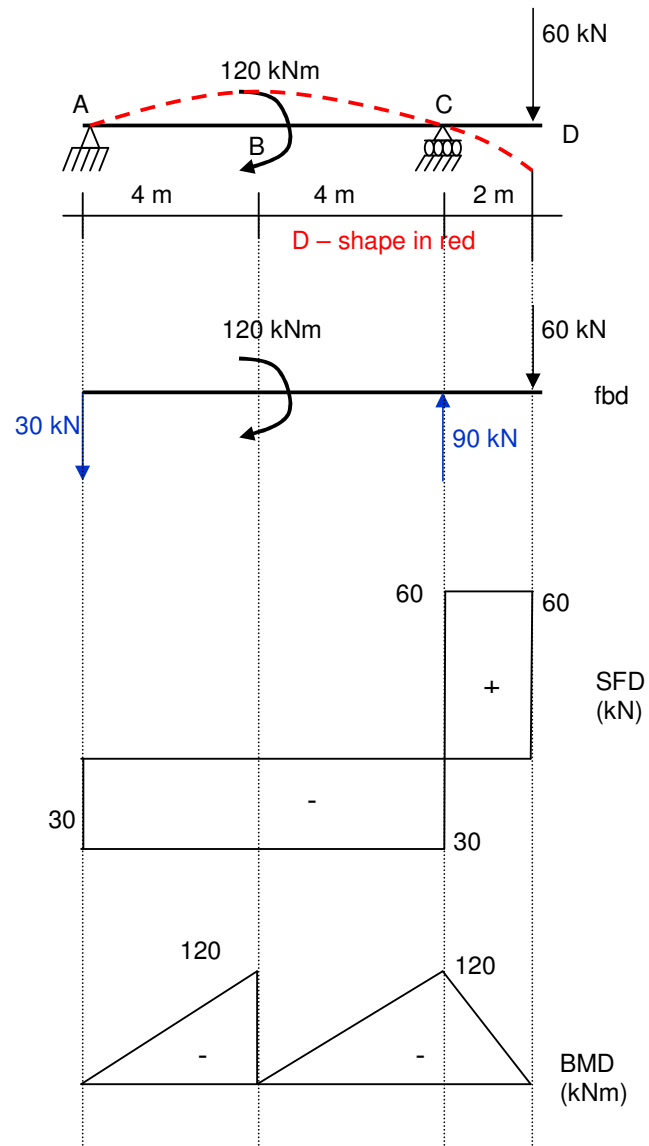
Example 6



Example 7



Example 8



5.5 ANALYSIS OF FRAMES

A statically determinate and open plane frame may be imagined to be “spread out” from the left hand side to the right hand side as though it were a continuous beam. The same convention may then be used to plot the AFD, SFD, BMD and sketch the deflected shape. The base line changes as each member orientation changes. The concept of “above” and “below” the baseline must therefore be applied in a consistent manner.

For an open 2-D frame, $\eta = r - (3 + i)$

For a closed 2-D frame with fixed supports, $\eta = r + 3m - 3j$

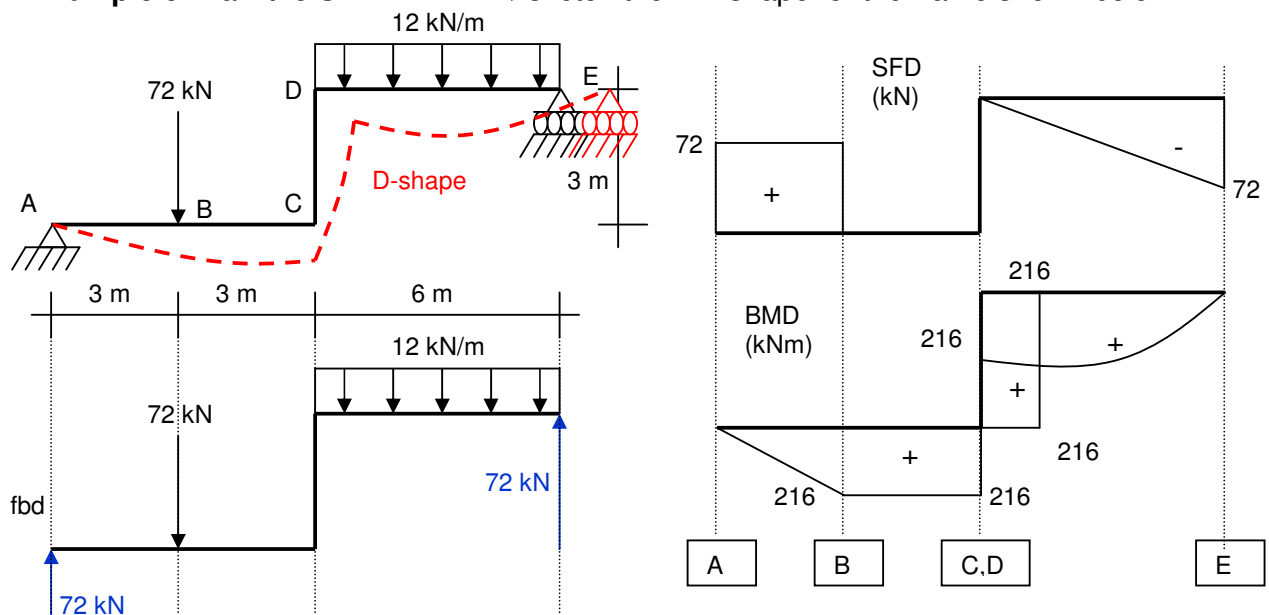
[Closed frames are statically indeterminate]

Rotational equilibrium at a rigid joint must be observed, thus:-

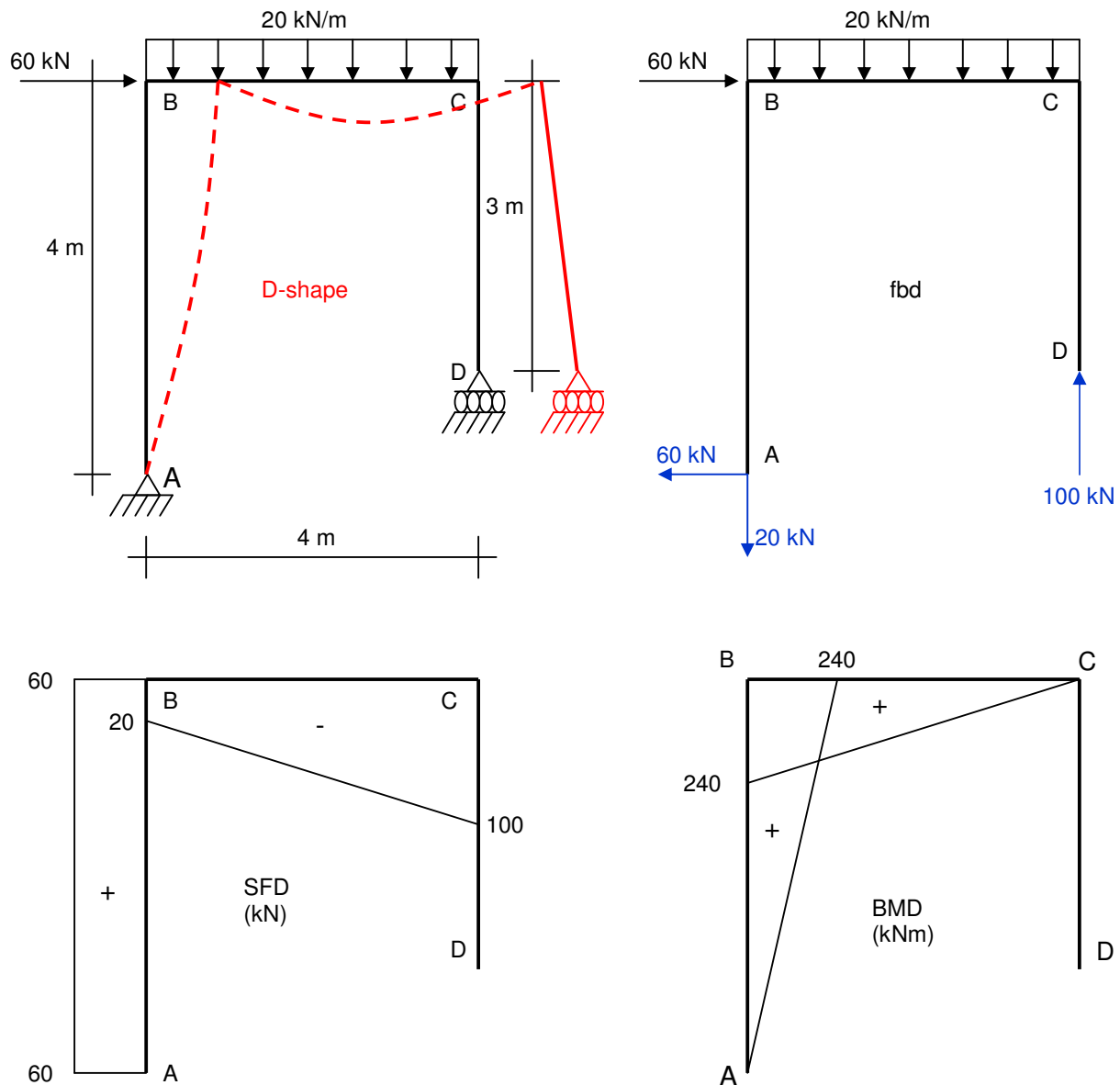
- ✓ The algebraic sum of the bending moments at the ends of all the members meeting at a rigid joint must be zero.
- ✓ If only two members meet at a rigid joint, the bending moments at the ends of the members must be equal in magnitude and opposite in sense.

In sketching the deflected shape, the original geometry of a rigid joint must be maintained. Deflections, as seen from a sketch of the deflected shape, are exaggerated for purposes of clarity. Actual deflections are very small. Axial strains are ignored.

Example 9 Draw the SFD, BMD and sketch the D – shape for the frame shown below.



Example 10 Draw the SFD, BMD and sketch the D – shape for the frame shown below.



The END of FCE 201